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Deep reinforcement learning assisted hyperheuristic for the large scale team orienteering problem with truck–drone cooperation

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ABSTRACT

Truck–drone collaborative information collection exploits the complementary capabilities of trucks and unmanned aerial vehicles (UAVs) to improve task performance under limited energy and task-duration constraints. This paper studies a truck–drone team orienteering problem (TD-TOP) for information collection tasks, to maximize the total collected information value under routing, synchronization, energy, and task-duration constraints. The proposed problem captures UAV sorties serving multiple candidate nodes, continuous truck progression, explicit launch, landing, and recovery operations, and bounded mutual waiting times between the truck and UAVs. A unified energy budget is adopted to jointly account for flight, service, and waiting activities within the same framework. A mixed-integer linear programming (MILP) model is first formulated to define the problem. To address this problem, a learning-assisted metaheuristic framework, termed PPO–VNS, is proposed by integrating Proximal Policy Optimization (PPO) with variable neighborhood search (VNS). PPO–VNS learns to select among heterogeneous neighborhoods to increase collected value while preserving feasibility. Computational results show that the proposed approach outperforms benchmark heuristics in collected information value across a wide range of test instances. On large-scale instances, it achieves average objective-value improvements of 5%–12% and identifies feasible solutions more reliably under tight constraints.

1. Introduction

The Team Orienteering Problem (TOP), also known as the Selective Traveling Salesman Problem (STSP), is a classical variant of the Vehicle Routing Problem (VRP) (Gunawan et al., 2016). It requires multiple vehicles to select and visit a subset of nodes within a limited time budget to maximize the total profit collected. It has been widely deployed in applications such as last-mile delivery, search and rescue, environmental monitoring, and time-critical information collection (Dursunoglu et al., 2025).

In such missions, unmanned aerial vehicles (UAVs), commonly referred to as drones, have been increasingly deployed due to their small size, high mobility, and low operating costs (Chen et al., 2025; Pasha et al., 2022). However, limited UAV battery capacity restricts flight range and the number of nodes served per sortie. Truck–drone cooperation mitigates this limitation by using the truck as a mobile launch-and-recovery base while UAVs serve nearby nodes (Chung et al., 2020; Garg et al., 2023). When this cooperative scheme is incorporated into the TOP, the resulting problem can be regarded as a truck–drone team orienteering problem (TD-TOP). The routes must jointly satisfy truck–drone synchronization, mission-duration constraints, and UAV energy constraints, making TD-TOP significantly more complex than the classical TOP. Unlike the classical TOP, TD-TOP requires coupled routing and scheduling decisions for heterogeneous vehicles. The truck route determines where and when UAVs can be launched and recovered, while UAV sorties affect truck waiting time, recovery feasibility, and UAV energy consumption. In addition, each UAV may visit multiple candidate nodes within a single sortie. Thus, sortie feasibility depends on node selection, visit order, service time, waiting time, and energy use. Therefore, point-of-interest selection, truck movement, UAV routing, launch and recovery operations, synchronization, and energy feasibility must be optimized jointly. Only a few studies have formulated truck–drone cooperation within a team orienteering framework. Zhang et al. (2021) consider a one-truck-one-drone scheme for search-and-rescue missions. Morandi et al. (2024) develop a one-truck-multi-drone scheme for last-mile delivery, in which each UAV visits a single point per sortie. These studies provide important extensions of the TOP under the emerging application of UAVs. However, existing studies on truck–drone team orienteering are still primarily developed for search-and-rescue or delivery-oriented settings, which do not fully capture information-collection operations

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under limited time and energy budgets. To the best of our knowledge, truck–drone cooperative TOPs tailored to information-collection operations under limited time and energy budgets have not yet been sufficiently investigated.

Motivated by this gap, the paper investigates TD-TOP for information-collection operations. Candidate nodes carry fixed information values and are visited selectively rather than mandatorily. Moreover, a truck may coordinate multiple UAVs, and each UAV may visit multiple candidate nodes within a single sortie under coupled energy and operation-duration constraints. Compared with one-truck-one-UAV settings, using multiple UAVs introduces additional coordination requirements because several UAVs may be launched, recovered, or remain in flight during overlapping time intervals. Compared with single-visit settings, the multi-visit structure requires each sortie to specify launch and recovery points, selected candidate nodes, and visit order. Consequently, the feasibility of a UAV sortie becomes sequence-dependent, since flight time, service time, waiting time, and energy consumption are accumulated along the entire UAV path. These features create a much tighter coupling among point-of-interest selection, routing, scheduling, synchronization, and energy feasibility than in delivery-oriented schemes. Therefore, existing delivery-oriented models and algorithms cannot be directly applied to the proposed information-collection problem without substantial modifications.

To address this challenge, heuristic and metaheuristic methods are often adopted to balance solution quality and computational cost. In large-scale, tightly constrained instances, heuristic performance depends strongly on how neighborhoods are selected during search. Fixed, random, or manually tuned rules often fail to adapt to changing solution states. As a result, these methods may suffer from inefficient exploration and limited search performance.

Recent learning-assisted neighborhood search methods, including bandit-based neighborhood selection (Kalatzantonakis et al., 2023), DQN-based operator selection (Zhou et al., 2026b), learning-assisted adaptive large neighborhood search (ALNS) (Kallestad et al., 2023), and neural improvement heuristics (Drakulic et al., 2023), provide useful references for adaptive search control. Building on this line of research, we develop a deep reinforcement learning (DRL)-based framework to guide the selection of neighborhood operators for TD-TOP. The neighborhood-operator selection process within VNS is formulated as a Markov decision process (MDP), where the trained policy serves as an adaptive operator-selection framework that determines which neighborhood operator to apply based on the current solution state and search progress.

Based on this idea, we propose a hybrid framework, termed PPO–VNS, that integrates Proximal Policy Optimization (PPO) with Variable Neighborhood Search (VNS). The contribution of PPO–VNS does not lie in learning-based operator selection alone, but in the integration of Recurrent PPO with a TD-TOP-specific VNS framework. Unlike fixed-rule or random operator-selection strategies, PPO–VNS selects operators according to multidimensional search-state information. Compared with standard value-based operator selection without recurrent memory, the Long Short-Term Memory (LSTM) structure embedded in Recurrent PPO enables the policy to learn how the effectiveness of an operator depends on previously selected operators and the recent search trajectory. Rather than constructing complete solutions end-to-end, PPO–VNS uses reinforcement learning to guide neighborhood selection. At the same time, TD-TOP-specific operators and feasibility checking handle multi-UAV coordination, multi-node UAV sorties, truck–drone synchronization, bounded waiting, and unified energy feasibility. In this way, PPO–VNS supports more adaptive and effective search in large-scale truck–drone information collection instances. The main contributions of this paper are summarized as follows.

(1) This paper introduces TD-TOP for information collection operations under limited time and energy budgets, which extends the classical team orienteering scheme to a coordinated truck–drone scenario with profit-driven candidate node selection. The proposed problem explicitly captures multi-node UAV sorties, continuous truck movement during UAV operations, coordinated launch and recovery, and joint time-energy feasibility.

(2) This paper develops a hybrid algorithm, termed PPO–VNS, for large-scale TD-TOP instances in information collection operations. The proposed method combines 15 problem-specific neighborhood operators, designed to reflect the structural characteristics of TD-TOP, with a PPO-based operator-selection mechanism embedded within a VNS framework. In this way, operator selection is transformed from a manually designed search component into a learned adaptive decision process, thereby improving the adaptability of neighborhood search.

(3) Extensive computational experiments are conducted on instances with 100–1000 nodes to evaluate the proposed framework. The results show that PPO–VNS obtains higher collected information values and shorter online computation times than the benchmark methods, especially on large and tightly constrained instances. In addition, the experimental analysis provides useful insights into truck–drone coordination strategies for information collection planning.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 analyzes the problem and presents a mathematical model; Section 4 details the solution algorithm; Section 5 reports and analyzes the experimental results; and Section 6 discusses the conclusions and future research directions.

2. Related literature

The classical TOP considers a fleet of vehicles that select and visit a subset of profitable nodes under limited travel time or capacity, with the aim of maximizing the total collected profit (Vansteenwegen et al., 2009). Many studies enrich the classical TOP with application-driven constraints and propose a broad spectrum of solution paradigms, including exact mathematical programming, scalable heuristics, and learning-assisted frameworks (Dursunoglu et al., 2025). The Capacitated Team Orienteering Problem (CTOP) extends TOP by adding vehicle capacity constraints (Luo et al., 2013): Angelo et al. (2024) address CTOP with heuristic frameworks, while Dursunoglu et al. (2025) develops a hybrid heuristic-exact scheme. In the Team Orienteering Problem with Time Windows (TOPTW), each selected node must be served within its specified time window (Hu & Lim, 2014). Li et al. (2024) and Peng et al. (2025) solve the problem using exact branch-and-price and branch-and-cut methods, while Long and Long (2024) and Zhou et al. (2026a) rely on heuristic frameworks such as adaptive large neighborhood search (ALNS) and local search. The Dynamic Team Orienteering Problem (DTOP) is a TOP variant in which problem information changes over time, requiring online route updates (Kirac et al., 2025). Paradiso et al. (2025) addressed TOP with time-varying profits via non-uniform distribution randomization. Neroni et al. (2025) solved TOP with stochastic travel times using heuristics combined with Monte Carlo simulation; and Kirac et al. (2025) proposed an online rolling-plan approach maintained by ALNS. In recent years, TOP has been introduced into UAV route planning and task allocation. Bayliss et al. (2020) extended the TOP to multi-UAV scenarios and addressed the UAV task allocation problem using metaheuristic algorithms. Sundar et al. (2022) and Fuertes et al. (2023) used exact algorithms to solve the resulting multi-UAV TOP model. In addition, several studies integrated machine learning with evolutionary or neighborhood-search methods, such as combining deep reinforcement learning with genetic algorithms (Vathis et al., 2024) or coupling Q-learning with ALNS (Lin et al., 2025) to guide the search process adaptively.

The truck–drone coordination problem was first proposed by Murray and Chu (2015) through the Flying Sidekick Traveling Salesman Problem (FSTSP), which assumes that a UAV can serve at most one customer per flight and focuses on the synchronized scheduling between truck routes and UAV flights. The study formulated an MILP model and developed a scalable route-and-reassign heuristic. Subsequently, Agatz et al. (2018) developed the Traveling Salesman Problem with Drone (TSP-D)

integer programming model, allowing trucks to launch UAVs multiple times from the same node, and proposed route-first-cluster-second solution schemes based on tour partitioning and local improvement. Tu et al. (2018) extended the model to a scheme in which one truck carries multiple UAVs, termed the Traveling Salesman Problem with multiple Drones (TSP-mD). This expansion increased synchronization and decision complexity. They designed the Greedy Randomized Adaptive Search Procedure and the ALNS metaheuristics. Murray and Raj (2020) further generalized FSTSP to the multi-FSTSP (mFSTSP) with heterogeneous UAVs in terms of flight speed, payload capacity, and energy constraints. However, it still restricts each flight to serving only one customer at a time and limits the truck to having only one UAV taking off or landing at any given moment. They provided an MILP formulation and employed a decomposition-style heuristic that solves a sequence of subproblems to obtain high-quality solutions at practical scales. Wang et al. (2017) extended the problem to multiple vehicles, formulating the Vehicle Routing Problem with Drones (VRP-D), and derived tight worst-case bounds on the achievable time savings as functions of k and the UAV-to-truck speed ratio α. Beyond these canonical models, recent studies have investigated richer operational assumptions, including two-echelon truck–drone routing with multi-visit service and load-dependent energy consumption (Liu et al., 2021), wind-affected drone-truck routing with nonlinear energy consumption under wind-speed and wind-direction uncertainty (Dodge et al., 2024), and dynamic truck–drone collaboration with flexible takeoff and landing truck assignment and multiple synchronization constraints (Zhao et al., 2025). In addition, learning-based decision models have been explored for truck–drone coordination, including attention-based encoder-decoder frameworks that learn to construct coordinated truck and UAV tours (Wu et al., 2023), uncertainty-aware settings where policy and value function approximation support dynamic execution decisions under travel-time uncertainty (Steenbergen et al., 2023), multi-agent deep reinforcement learning for truck–drone coordination with mobile charging (Qiu et al., 2026), and a reinforcement learning framework developed as a scalable alternative to an exact MILP benchmark for traffic-weighted hybrid truck–drone delivery with deadlines (Shuaibu et al., 2026). Overall, these studies establish core modeling and algorithmic foundations for truck–drone coordination, yet they largely focus on delivery-type objectives (Agatz et al., 2018; Dodge et al., 2024; Eker et al., 2025; Liu et al., 2021; Luo et al., 2021; Maini et al., 2019; Murray & Chu, 2015; Murray & Raj, 2020; Qiu et al., 2026; Shuaibu et al., 2026; Steenbergen et al., 2023; Tu et al., 2018; Wang et al., 2017; Wu et al., 2023; Zhao et al., 2025). A comparative summary of representative TOP variants and truck–drone coordination studies is provided in Table 1.

For information collection, TOP naturally models selective UAV visits under duration and energy limits (Moskal et al., 2023). However, limited onboard endurance restricts coverage per sortie, motivating truck–drone cooperation in which a ground vehicle serves as a mobile base for launch and recovery. This cooperative setting introduces spatiotemporal coupling constraints that are not captured by conventional TOP variants. Only a few works explicitly cast truck–drone cooperation within an orienteering scheme. Zhang et al. (2021) modeled post-disaster assessment as a Combined Drone Orienteering Problem (CDOP), formulated it as an MILP, and proposed a column-generation-based heuristic. Morandi et al. (2024) introduced the Orienteering Problem with multiple Drones (OP-mD), in which a truck cooperates with several UAVs to maximize the total collected profit within a time budget, and solved it using a tailored MILP and a branch-and-cut algorithm. These pioneering studies demonstrate the suitability of orienteering objectives for emerging UAV applications. Yet, they primarily target emergency response or delivery-related scenarios rather than truck–drone information collection operations under limited time and energy budgets (Grocholsky et al., 2006; Luo et al., 2018; Sun et al., 2024). More importantly, they do not address the combined scheme of information-value-driven candidate-node selection, moving-truck support, multi-node UAV sorties, and tightly coupled time–energy feasibility.

Recently, reinforcement learning (RL) has been increasingly applied to routing and scheduling problems (Peng et al., 2023; Yuan et al., 2025) providing data-driven and adaptive solution paradigms. Existing work can be broadly categorized into construction-based and improvement-based approaches (Zhou et al., 2025). Vinyals et al. (2015) and Nazari et al. (2018) proposed pointer-network and attention-based models for end-to-end route construction, while Kool et al. (2019), Jin et al. (2023), Yang et al. (2023), and Lischka et al. (2024) adopted Transformer variants to enhance scalability and generalization. On the improvement side, Chen and Tian (2019) developed NeuRewriter for rewrite-based local search, Ma et al. (2023) learned generalized k-opt actions, Kalatzantonakis et al. (2023) used bandit-based adaptive neighborhood selection in VNS, Hottung and Tierney (2020) and Yang et al. (2023) designed neural repair operators for LNS, and Zhou et al. (2026b) combined DQN with ALNS to learn destroy and repair operator selection. Despite this progress, learning-guided improvement paradigms have been far less explored for truck–drone cooperative schemes with strict synchronization constraints, where the search landscape is shaped not only by routing choices but also by coupled launch-recovery timing, multi-platform coordination, and feasibility under shared time and energy budgets. These characteristics motivate the proposed DRL-assisted hyperheuristic framework to support scalable search for large-scale truck–drone cooperative team orienteering in information collection tasks.

Based on the above review and the comparative analysis in Table 1, several research gaps can be summarized as follows.

1) Existing TOP variants have provided solid foundations for selective node visitation and profit-oriented routing. However, relatively limited attention has been paid to information collection tasks supported by a moving truck and multiple UAVs.

2) Existing truck–drone coordination studies are often formulated with objectives such as minimizing completion time or operational cost. In contrast, information collection tasks place greater emphasis on value-driven candidate-node selection under limited time and resources, leading to a different modeling focus.

3) Some studies have introduced orienteering objectives into truck–drone cooperation. Nevertheless, the joint consideration of moving-truck support, multiple UAVs, multi-node UAV sorties, and truck–UAV synchronization still leaves room for further investigation.

4) Learning-assisted routing methods have shown strong potential in combinatorial optimization. Yet, their application to truck–drone cooperative team orienteering with coupled routing, timing, and energy constraints remains relatively underexplored.

Motivated by these observations, this study formulateresourcess a truck–drone cooperative team orienteering problem for information collection and develops a DRL-assisted hyperheuristic framework to improve the search efficiency for large-scale instances.

3. Problem description and mathematical model

This section describes TD-TOP for information collection tasks in a one-truck multi-UAV scheme and presents its MILP model.

3.1. Problem description

Let $V_1 = \{0, \dots, l\}$ be the truck-accessible road nodes, with node 0 as the depot where the truck starts and ends its route. The UAV fleet is initially carried by the truck when it departs from node 0 and is finally returned to the depot together with the truck after the operation. Thus, node 0 represents the initial and final storage location of the UAV fleet, rather than the fixed launch and recovery node for every UAV sortie. Let $V_2 = \{l+1, \dots, n\}$ denote the set of candidate information-collection nodes. Each node $i \in V_2$ is associated with a fixed information value p_i, which corresponds to the profit parameter in the classical TOP. Therefore, $V = \{V_1 \cup V_2\}$ represents the union of all nodes in the graph. TD-TOP in a one-truck multi-UAV scheme is defined on a graph (V, E), where $E = E_1 \cup E_2$ gives the edge set. Specifically, $E_1 = \{(i, j) \mid$

Table 1
Summary of representative TOP variants and truck–drone coordination studies.

Article(s)	problem	vehicle	UAV	multi visit
Angelo et al. (2024), Dursunoglu et al. (2025)	CTOP	multiple vehicles	–	✓
Hu and Lim (2014), Li et al. (2024), Long and Long (2024), Peng et al. (2025), Zhou et al. (2026a)	TOPTW	multiple vehicles	–	✓
Kirac et al. (2025), Neroni et al. (2025), Paradiso et al. (2025)	DTOP	multiple vehicles	–	✓
Bayliss et al. (2020), Fuertes et al. (2023), Sundar et al. (2022)	TOP	–	multiple drones	✓
Murray and Chu (2015)	FSTSP	1 truck	1 UAV	–
Agatz et al. (2018)	TSP-D	1 truck	1 UAV	–
Tu et al. (2018)	TSP-mD	1 truck	multiple UAVs	–
Murray and Raj (2020)	mFSTSP	multiple trucks	multiple UAVs	–
Wang et al. (2017)	VRP-D	multiple trucks	1 UAV per truck	–
Liu et al. (2021)	VRP	multiple trucks	multiple UAVs	✓
Dodge et al. (2024)	DTCO/FSTSP	1 truck	1 UAV	✓
Zhao et al. (2025)	DTDC	multiple trucks	multiple UAVs	–
Wu et al. (2023)	VRP-D	multiple trucks	1 UAV per truck	–
Zhang et al. (2021)	CDOP	1 truck	1 UAV	✓
Morandi et al. (2024)	OP-mD	1 truck	multiple UAVs	–
This paper	TD-TOP	1 truck	multiple UAVs	✓

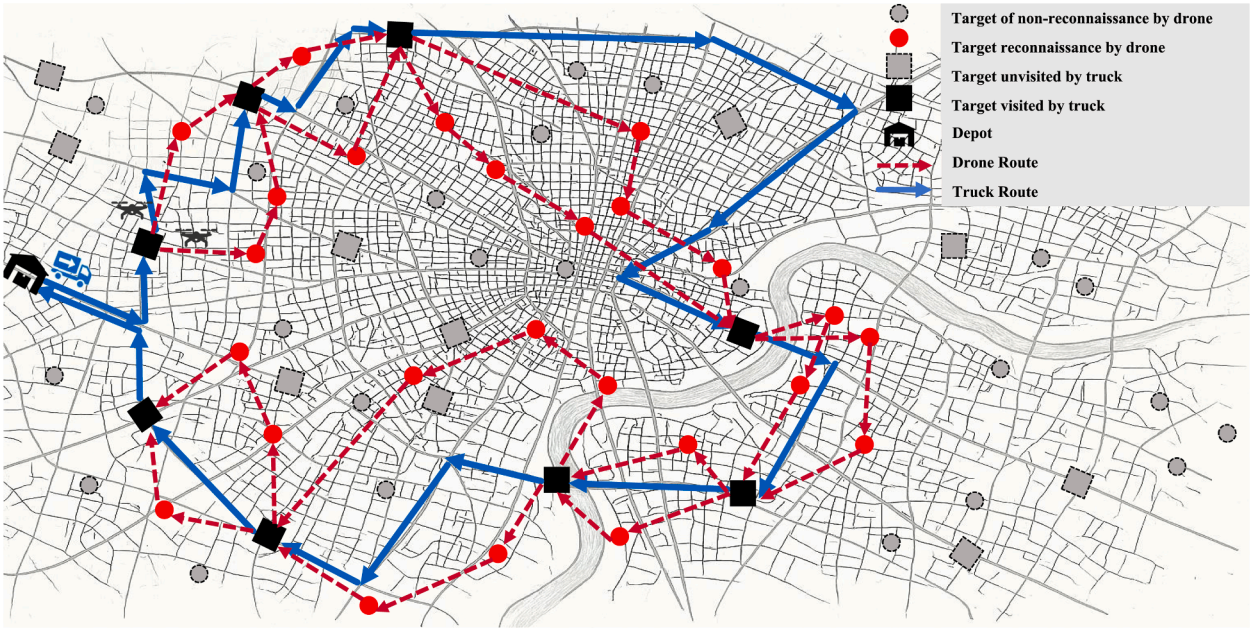


Fig. 1. TD-TOP for information collection tasks.

$i, j \in V_1,\ i \neq j\}$ denotes the set of road-network edges traversable by the truck, and $E_2 = \{(i,j) \mid i, j \in V,\ i \neq j\}$ denotes the set of feasible flight edges available to the UAVs for information collection operations, as shown in Fig. 1.

The truck carries up to R identical UAVs for information collection tasks. Each candidate node can be visited at most once. The service time required by a UAV at candidate node $i \in V_2$ is denoted by s_i. At any road-network node visited by the truck, one or more UAVs may be launched or recovered. Each UAV departs from a truck-visited road-network node and is recovered at another truck-visited road-network node. The truck does not collect information by itself, but serves as a mobile launch-and-recovery platform.

Let K denote the set of UAV sorties. For any $(i,j) \in E_1$, the truck travel time is denoted by t_{ij}^G, and for any $(i,j) \in E_2$, the UAV travel time is denoted by t_{ij}^U. In the computational experiments, t_{ij}^G is calculated as $d_{ij}^{\mathrm{Man}}/v^G$, where d_{ij}^{Man} is the Manhattan distance between nodes i and j and v^G is the truck speed. Similarly, t_{ij}^U is calculated as $d_{ij}^{\mathrm{Euclid}}/v^U$, where d_{ij}^{Euclid} is the Euclidean distance and v^U is the UAV speed.

The truck is assumed to have no endurance limit. During the information collection operation, each UAV consumes energy for flight, service, and hover waiting. Each UAV sortie consists of a launch node, one or more candidate nodes, and a recovery node, and must satisfy the battery-capacity constraint θ. If a UAV is not assigned to a sortie, it remains on the truck and moves with the truck. Launch, landing, recovery, and battery replacement times are assumed to be negligible. This assumption is made because the proposed model focuses on mission-level route planning, where truck travel time, UAV flight time, service time, and hover waiting time are the dominant time components. Battery replacement is assumed to be performed on the truck using spare batteries prepared in advance. If these operation times are non-negligible in a specific application, they can be incorporated as additional fixed operation times at the corresponding launch and recovery nodes. When the dispatched UAV arrives at the recovery node before the truck, it is allowed to hover and wait at the node until the truck arrives. During this waiting period, the UAV continues to consume energy and must be recovered by the truck before exhausting its remaining battery. Multiple UAVs may be recovered at different times. However, all UAVs assigned to depart from

the same truck stop must be launched simultaneously before the truck leaves the node.

3.2. UAV endurance capability

Rotary-wing UAVs are well-suited to information collection operations due to their hovering and maneuvering capabilities. Their energy consumption consists of three components: flight, hover waiting, and information-collection service. Accordingly, the total energy consumption is modeled as the sum of these three parts.

For a rotary-wing UAV, the flight power is calculated as follows (Zeng et al., 2019):

$$P^{fly}(V) = P_0\left(1+\frac{3V^2}{U_{tip}^2}\right) + P_i\left(\sqrt{1+\frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho q A V^3 \tag{1}$$

where P_0 and P_i are the blade profile power and hover-induced power, respectively. U_{tip} denotes the rotor tip speed, v_0 denotes the induced velocity in hover, d_0 denotes the fuselage drag coefficient, ρ denotes the air density, q denotes the rotor solidity, and A denotes the rotor disk area.

The truck is assumed to have no effective endurance limit within the planning horizon. During task execution, each UAV consumes energy in three forms: flight, hover waiting, and information-collection service. Since UAV batteries can be replaced on the truck after recovery, the battery-capacity constraint is imposed at the level of each UAV sortie. Battery replacement is assumed to take negligible time.

Since the takeoff and landing phases are assumed to be negligible relative to the cruise phase, each UAV flight segment is modeled as constant-speed motion. Accordingly, the energy consumption on flight arc (i, j) is expressed as

$$E_{ij}^{fly} = P^{fly}(v^U)t_{ij}^U, \tag{2}$$

where t_{ij}^U denotes the UAV flight time on arc (i, j), and v^U denotes the UAV flight speed.

If a UAV reaches its recovery node before the truck arrives, it may hover and wait for recovery. The corresponding waiting energy consumption is expressed as

$$E_k^{wait} = P^{hover}t_k^{U,W} = P^{fly}(0)t_k^{U,W} = (P_i + P_0)t_k^{U,W}, \tag{3}$$

where $t_k^{U,W}$ denotes the hover-waiting time of the k-th UAV sortie, and P^{hover} denotes the hovering power of the UAV.

During the information-collection service at node i, the UAV consumes additional power for sensing, onboard computing, and communication, in addition to the power required for hovering. The corresponding service energy consumption is expressed as

$$E_i^{srv} = P^{srv}s_i = (P^{avio} + P^{hover})s_i, \tag{4}$$

with

$$P^{avio} = P^{sens} + P^{comp} + P^{tx}, \tag{5}$$

where P^{avio} denotes the total power consumption of the information-collection equipment during the service process (Meskar & Ahmadi-Javid, 2026). Here, P^{sens}, P^{comp}, and P^{tx} denote the power consumption of the sensor, onboard computing, and communication modules, respectively, and s_i denotes the information-collection service time at node i.

Combining the three components above, the total energy consumption of the k-th UAV sortie is expressed as

$$E_k = \sum E_{ij}^{fly} + \sum E_i^{srv} + E_k^{wait}, \tag{6}$$

where the summations are taken over the flight arcs and candidate nodes associated with sortie k.

Table 2
Definitions of the decision variables in the proposed MILP model.

Decision variable	Description
$y_{ij} \in \{0,1\}$	Determine whether the truck travels along the arc (i, j)
$z_i^G \in \{0,1\}$	Determine whether node i is visited by the truck
$z_{i,k}^U \in \{0,1\}$	Determine whether node i is served on the kth flight route
$t_i^{G,A} \geq 0$	Arrival time of the truck at node i
$t_i^{G,L} \geq 0$	Departure time of the truck from node i
$0 \leq r_i \leq R$	Number of UAVs carried by the truck when it departs from road-network node i
$x_{ij,k} \in \{0,1\}$	Determine whether the edge $(i, j) \in E$ is included in the flight path of the kth UAV
$t_k^{U,A} \geq 0$	Arrival time of the UAV at the recovery node in sortie k
$t_k^{U,L} \geq 0$	Launch time of the UAV from the launch node in sortie k
$t_k^{U,W} \geq 0$	Hover-waiting time of the UAV in sortie k
$h_{i,k}^L \in \{0,1\}$	Determine whether node i serves as the launch node for the kth flight path
$h_{i,k}^R \in \{0,1\}$	Determine whether node i serves as the recovery node for the kth flight path
$\delta_k \in \{0,1\}$	Determine whether the UAV needs to wait for the truck in the kth UAV flight path
$\mu_i^G \geq 0$	Visiting order of road-network node i in the truck route
$\mu_{i,k}^U \geq 0$	Visiting order of node i in the kth UAV flight route

3.3. Decision variables and objectives

The objective of the TD-TOP involving one truck and multiple UAVs is to maximize the total information value collected within the specified operation duration T, while ensuring that both the UAVs and the truck return to the depot. The main decision variables are summarized in Table 2.

If the index $k \in K$ is not used, then the decision variable associated with that index should be set to 0. Additionally, the integer decision variable r_i $(i \in V)$ represents the number of UAVs carried by the truck upon departing node i.

The binary variable $z_{i,k}^U$ equals 1 if candidate node $i \in V_2$ is served by a UAV in sortie k, and 0 otherwise. Similarly, the binary variable $x_{ij,k}$ equals 1 if arc $(i, j) \in E$ is traversed by a UAV in sortie k, and 0 otherwise. The variables $z_{i,k}^U$, $x_{ij,k}$, $h_{i,k}^L$, and $h_{i,k}^R$ jointly define the structure of UAV sortie k: $z_{i,k}^U$ identifies the served candidate nodes, $x_{ij,k}$ determines the selected flight arcs, and $h_{i,k}^L$ and $h_{i,k}^R$ specify the launch and recovery nodes, respectively.

For UAV sortie k, the launch time from the launch node and the arrival time at the recovery node are denoted by $t_k^{U,L}$ and $t_k^{U,A}$, respectively. Since the flight time can be obtained from the sum of t_{ij}^U over the traversed flight arcs and the service time can be obtained from the sum of s_i over the served candidate nodes, the corresponding hover-waiting time can be determined accordingly. A binary auxiliary variable δ_k is introduced to indicate whether hover waiting occurs in sortie k, and $t_k^{U,W}$ denotes the associated waiting time.

Finally, the auxiliary variables μ_i^G and $\mu_{i,k}^U$ denote the visiting order of node i in the truck route and in the kth UAV flight route, respectively.

3.4. Mathematical model

Basic path constraints based on flight segment modeling:

$$\max \sum_{i\in V_2}\sum_{k\in K} p_i z_{i,k}^U \tag{7}$$

$$\text{s.t.} \quad \sum_{j\in V} x_{ji,k} \geq z_{i,k}^U, \forall i \in V_2,\ \forall k \in K \tag{8}$$

$$h_{i,k}^R + \sum_{j\in V} x_{ij,k} = h_{i,k}^L + \sum_{j\in V} x_{ji,k}, \forall i \in V,\ \forall k \in K \tag{9}$$

$$\sum_{j\in V} x_{ij,k} \le 1, \forall i \in V,\ \forall k \in K \tag{10}$$

$$\sum_{j\in V} x_{ji,k} \le 1, \forall i \in V,\ \forall k \in K \tag{11}$$

$$\sum_{j\in V_1} y_{ij} = \sum_{j\in V_1} y_{ji} \le 1, \forall i \in V_1 \tag{12}$$

$$\sum_{k\in K} x_{ij,k} + \sum_{k\in K} x_{ji,k} \le 1, \forall (i,j) \in E \tag{13}$$

$$z^U_{i,k} + z^U_{j,k} \ge 1 - M(1 - x_{ij,k}), \forall i,j \in V_2,\ i \ne j,\ \forall k \in K. \tag{14}$$

$$h^L_{i,k} \ge 1 - M\left(2 - x_{ij,k} - z^U_{j,k} + z^U_{i,k}\right), \quad \forall i \in V_1,\ \forall j \in V_2,\ i \ne j,\ \forall k \in K \tag{15}$$

$$h^R_{j,k} \ge 1 - M\left(2 - x_{ij,k} - z^U_{i,k} + z^U_{j,k}\right), \quad \forall i \in V_2,\ \forall j \in V_1,\ i \ne j,\ \forall k \in K \tag{16}$$

$$\sum_{k\in K} h^L_{i,k} \le M \sum_{j\in V} y_{ij}, \forall i \in V_1 \tag{17}$$

$$\sum_{k\in K} h^R_{i,k} \le M \sum_{j\in V} y_{ji}, \forall i \in V_1 \tag{18}$$

$$h^L_{i,k} + h^R_{i,k} \le 1, \forall i \in V_1,\ \forall k \in K \tag{19}$$

$$h^L_{i,k} + h^R_{j,k} \le 1 + y_{ij}, \forall (i,j) \in E_1,\ \forall k \in K \tag{20}$$

$$\sum_{i\in V_1} y_{0i} = 1, \qquad \sum_{i\in V_1} y_{i0} = 1 \tag{21}$$

$$\sum_{k\in K} z^U_{i,k} \le 1, \forall i \in V_2 \tag{22}$$

$$1 \le \mu^G_i \le |V_1| - 1, \quad \forall i \in V_1 \setminus \{0\} \tag{23}$$

$$\mu^G_j \ge \mu^G_i + 1 - M(1 - y_{ij}), \forall i,j \in V_1 \setminus \{0\},\ i \ne j \tag{24}$$

$$0 \le \mu^U_{i,k} \le |V| - 1, \quad \forall i \in V,\ \forall k \in K \tag{25}$$

$$\mu^U_{j,k} \ge \mu^U_{i,k} + 1 - M(1 - x_{ij,k}), \forall i,j \in V,\ i \ne j,\ \forall k \in K \tag{26}$$

Objective (7) maximizes collected information value. Constraint (8) ensures that a candidate node can be served by a UAV in sortie k only if there is at least one incoming UAV arc to that node. Constraint (9) enforces flow balance for each UAV sortie by accounting for the launch and recovery indicators together with the incoming and outgoing UAV arcs at every node. Constraints (10) and (11) restrict the out-degree and in-degree of each node in every UAV sortie to at most one. Constraint (12) imposes flow balance on the truck route and ensures that each road-network node is visited by the truck at most once. Constraint (13) prevents UAV traversal of both directions of the same flight edge. Constraint (14) requires that if a UAV traverses flight arc (i,j) in sortie k, then at least one endpoint of that arc must be selected for information collection in the same sortie. Constraints (15) and (16) determine the launch node and the recovery node of each UAV sortie, respectively. Specifically, if a traversed arc connects a road-network endpoint to a candidate information-collection endpoint, then the road-network endpoint is identified as the launch or recovery node based on the flight direction. Constraints (17) and (18) ensure that UAV launch and recovery can occur only at nodes visited by the truck. Constraint (19) requires that the launch node and the recovery node of a UAV sortie be different. Constraint (20) restricts UAVs to be launched and recovered at adjacent road-network nodes along the truck route. Finally, Constraint (21) ensures that the truck departs from and returns to the depot exactly once. Constraint (22) ensures that each candidate information-collection node can be served by at most one UAV sortie, thereby preventing repeated information collection at the same node. Constraints (23)-(24) and Constraints (25)-(26) are MTZ-type subtour-elimination constraints for the truck route and UAV flight routes, respectively. If an arc is selected, the visiting order of the downstream node must be larger than that of the upstream node, thereby preventing disconnected subtours.

$$r_0 + \sum_{j\in V_2} \sum_{k\in K} x_{0j,k} = R \tag{27}$$

$$r_i + \sum_{l\in V} \sum_{k\in K} x_{li,k} \ge r_j + \sum_{l\in V} \sum_{k\in K} x_{jl,k} - M(1 - y_{ij}), \quad \forall i,j \in V,\ i \ne j \tag{28}$$

$$r_i + \sum_{j\in V} \sum_{k\in K} x_{j0,k} \ge R - M(1 - y_{i0}), \forall i \in V \tag{29}$$

$$\sum_{i\in V} \sum_{j\in V} P^{fly} t^U_{ij} x_{ij,k} + \sum_{i\in V_2} P^{srv} s_i z^U_{i,k} + P^{hover} t^{U,W}_k \le \theta, \quad \forall k \in K \tag{30}$$

Constraints (27)–(29) ensure conservation of the total number of UAVs. Constraint (30) ensures that, for each UAV sortie, the sum of flight energy, information-collection service energy, and hover-waiting energy does not exceed the UAV battery capacity.

$$M(1 - y_{ij}) \ge |t^{G,A}_j - t^{G,L}_i - t^G_{ij}|, \forall (i,j) \in E \tag{31}$$

$$t^{G,L}_i \ge t^{U,A}_k - M(1 - h^R_{i,k}), \forall i \in V_1,\ \forall k \in K \tag{32}$$

$$M\left(1 - \sum_{i\in V} h^R_{i,k}\right) \ge |t^{U,A}_k - t^{U,L}_k - \sum_{i\in V} \sum_{j\in V} t^U_{ij} x_{ij,k} - \sum_{i\in V_2} s_i z^U_{i,k}|, \forall k \in K \tag{33}$$

$$M(1 - h^L_{i,k}) \ge |t^{U,L}_k - t^{G,L}_i|, \qquad \forall i \in V_1,\ \forall k \in K \tag{34}$$

$$t^{U,A}_k \le T, \forall k \in K \tag{35}$$

$$t^{G,A}_0 \le T \tag{36}$$

$$M(2 - \delta_k - h^R_{i,k}) \ge \left|t^{U,W}_k - \left(t^{G,A}_i - t^{U,A}_k\right)\right|, \quad \forall k \in K,\ i \in V_1 \tag{37}$$

$$M\delta_k \ge t^{U,W}_k, \forall k \in K \tag{38}$$

Constraint (31) enforces time propagation along the truck route. Constraint (32) requires that the truck can depart from a node only after the corresponding UAV recovery has been completed. Constraint (33) enforces time consistency for each UAV sortie. Constraint (34) ensures that a UAV takes off at the time when the truck departs from the corresponding launch node. Constraints (35) and (36) require that the return times of the UAVs and the truck to the depot do not exceed the task-duration time limit T. Constraints (37) and (38) determine the UAV hover-waiting time. For numerical stability, the big-M constants used in the MILP formulation are not assigned as a single excessively large value. Instead, constraint-specific upper bounds are used according to the variable ranges involved in each constraint. In particular, the values of M are determined based on the bounds of the associated binary variables, the number of UAV sorties, the maximum number of UAVs carried by the truck, and the operation duration T. This treatment avoids unnecessarily loose big-M constraints while preserving the validity of the formulation.

4. Solution algorithm

4.1. PPO–VNS framework

This section presents a hybrid algorithmic framework that combines PPO with VNS. To solve TD-TOP, a neighborhood selection framework based on Recurrent PPO is developed. The core idea is to model the neighborhood-operator selection process as a Markov decision process, in which the reinforcement learning agent selects an appropriate perturbation operator based on the current solution state and iteratively improves solution quality according to the collected information value. To enhance diversification and escape local optima, a

simulated-annealing (SA) acceptance criterion is incorporated, allowing non-improving moves to be accepted with a temperature-dependent probability according to the classical Metropolis rule (Kirkpatrick et al., 1983). In traditional VNS, neighborhood operators are typically selected randomly or with fixed probabilities, preventing the selection strategy from adapting to the current solution state. In the proposed framework, the agent employs an LSTM architecture to capture historical information. By memorizing and encoding the neighborhood-selection sequence and environment-state transitions from the previous iteration and earlier search history, the agent learns deeper temporal dependencies among operator actions. The pseudocode of the proposed algorithm is given in Algorithm 1.

4.2. Initial deconstruction

The initial solution is constructed through a clustering-based procedure. A hierarchical clustering strategy based on road-network nodes is adopted, in which each road-network node is taken as an initial cluster center, and the clustering radius is gradually expanded from zero until all candidate information-collection nodes are fully covered. The total information value of each road-network-node cluster is then computed. Next, a truck route with the depot as both the origin and destination is constructed by inserting road-network nodes in descending order of cluster information value until the overall task-duration limit is reached. Finally, for each pair of adjacent road-network nodes, candidate nodes within the corresponding cluster are greedily inserted according to information value priority until the UAV battery-capacity constraint or the overall task-duration limit is reached.

4.3. Neighborhood operator design

Four truck-route optimization operators and eleven UAV-route optimization operators are designed. Since changes in the truck route substantially impact the overall solution structure, truck operators are used as perturbation mechanisms to help the search escape local optima. Operator N_1 is a 2-opt operator, which randomly selects two nonadjacent edges in the truck route and reverses the subpath between them. Operator N_2 is a swap operator, which randomly removes one visited node from the truck route and inserts one unvisited node. Operator N_3 is an insert operator, which randomly inserts an unvisited node into the truck route. Operator N_4 is a delete operator, which randomly removes one node from the truck route. A schematic illustration of these operators is shown in Fig. 2.

To address the TD-TOP, eleven UAV-route optimization operators are designed for local search by jointly considering information value and energy consumption. Operator N_5 is an Ejection Chain operator. It inserts node 4 from route A into route B, then inserts node 7 from route B into route C. Through this chained relocation of nodes, the operator reduces waiting time across different routes in the road network. Operator N_6 is a Block Swap operator. It exchanges m consecutive nodes in route A with m consecutive nodes in route B, thereby exploring a larger neighborhood through block-level exchanges. This operator aims to reduce total flight energy consumption and compress the overall completion time. Operator N_7 is a Snap to Neighbor operator. It randomly selects a node from route A and inserts it into the route closest to that node. By identifying a nearby route based on geometric distance, this operator reduces flight distance, waiting time, and completion time. Operator N_8 is a Two-Opt-energy operator. Within the same route, it exchanges two nonadjacent edges and reverses the subpath between them, using energy reduction as the acceptance criterion. Operator N_9 is a Three-Opt-energy operator. Within the same route, it exchanges three nonadjacent edges and reverses the corresponding subpaths, again using energy reduction as the acceptance criterion. Operator N_{10} is an Or-Opt-srv(k) operator. Along the same route, it exchanges k nonadjacent edges and reverses the subpaths between them, using energy reduction as the acceptance criterion. Operator N_{11} is a Cross-Reverse operator. Between two different routes, it exchanges two nonadjacent edges and reverses the associated subpaths, using either reduced energy consumption or reduced waiting time as the acceptance criterion.

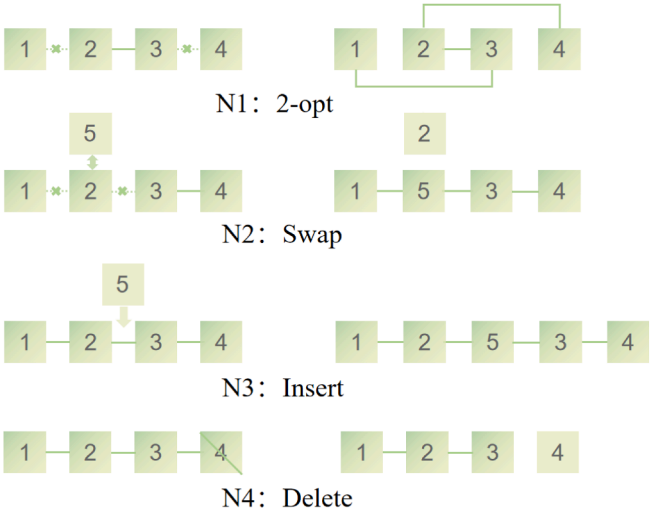


Fig. 2. Schematic diagram of truck neighborhood operators.

Operator N_{12} is a Relocate-energy operator. It randomly selects a node from a high-energy-consumption route segment and inserts it into a lower-energy-consumption route, thereby reducing global energy consumption and completion time. Operator N_{13} is a Drop-Low operator. It randomly removes the node with the lowest information value from a route segment to reduce energy consumption and waiting time. Operator N_{14} is an Insert-High operator. It selects a high-value node from the unvisited nodes, enumerates all feasible insertion positions, and chooses the one that yields the smallest increase in energy consumption, thereby improving the overall value of the collected information. Operator N_{15} is a Replace Low-High operator. It removes an already-visited low-value node and inserts an unvisited high-value node, thereby improving the overall value of the collected information. A schematic illustration of these operators is shown in Fig. 3.

4.4. Proximal policy optimization

Recurrent Proximal Policy Optimization is employed to select neighborhood operators for TD-TOP, and the operator-selection process is formulated as a Markov decision process (MDP). In this MDP, the agent learns, through interaction with the search environment, which operator to select in different search states to improve the overall objective. After training, the learned policy is embedded into the main loop of Variable Neighborhood Search and is used to guide local-search operator selection.

The MDP is defined by the four-tuple (S, A, P, R). Each state $s_t \in S$ is represented by a compact observation vector that characterizes the current search status, including solution quality, feasibility pressure, search dynamics, and operator-usage history. The observation consists of 20 scalar features and a one-hot encoding of the previously selected neighborhood operator. At each state s_t, the agent selects an operator $a_t \in A$ and receives an immediate reward r_t according to the reward function R. Improvements in solution quality and changes in time and energy consumption jointly determine the reward. After the selected operator is applied, the search environment is updated to s_{t+1} according to $P(s_{t+1} \mid s_t, a_t)$. The episode terminates within a finite number of operator-selection steps.

Unlike standard PPO, Recurrent PPO incorporates an LSTM structure into the policy network, enabling the agent to retain historical state information and capture temporal dependencies in the search trajectory. The policy is parameterized as $\pi_\theta(a_t \mid s_t, h_t)$, where h_t denotes the LSTM

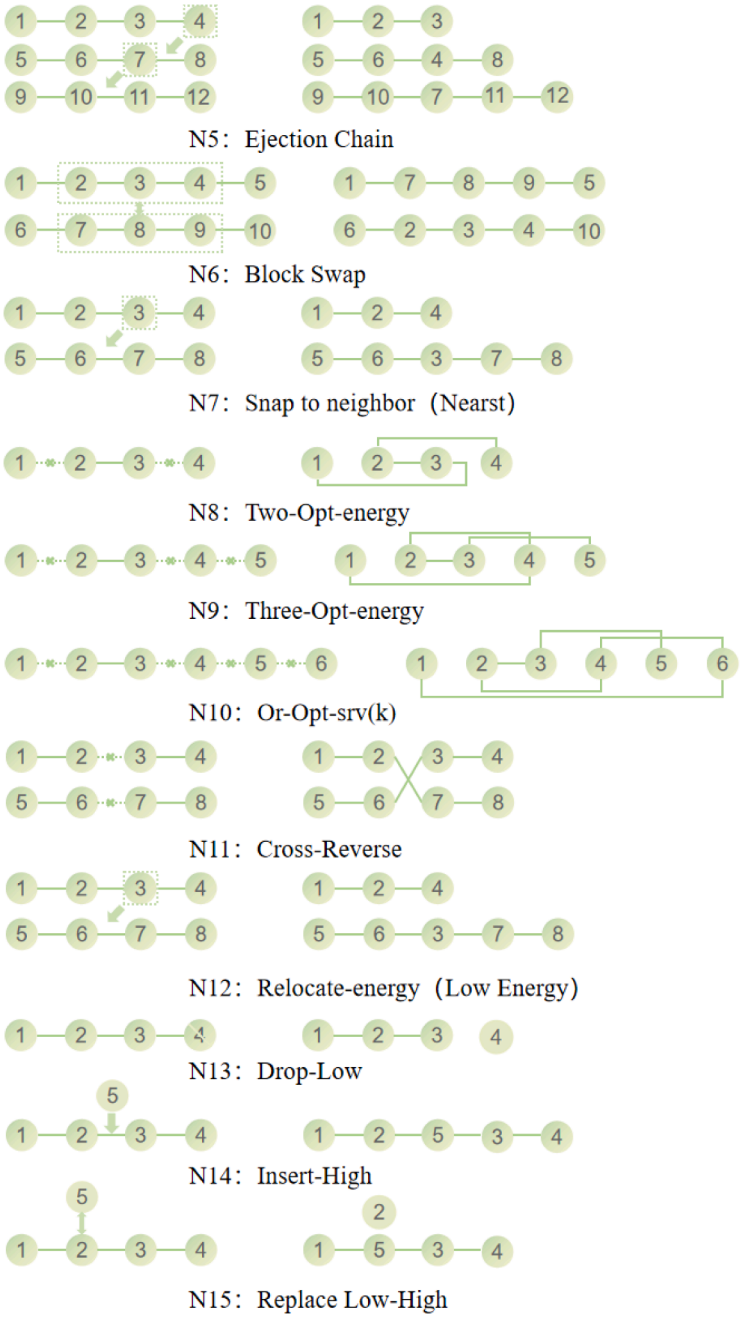


Fig. 3. Schematic diagram of UAV neighborhood operators.

hidden state. A value function $V_\phi(s_t, h_t)$ is used to estimate the state value, and the advantage $\hat{A}_t$ is computed for the PPO clipped surrogate objective. The previous-action one-hot vector provides an explicit short-term operator-history cue, whereas the LSTM hidden state captures longer-term search dependencies. The complete list of state-feature components is provided in Appendix A.1.

A compact state representation is designed for selecting neighborhood operators. The features summarize solution quality, feasibility-related slack information, search dynamics associated with the simulated-annealing mechanism, and operator usage history. In addition, the observation includes a one-hot encoding of the previous action to provide recent action context for the LSTM and mitigate short-term operator repetition. The complete feature list is provided in Appendix A.1.

4.4.1. Action

In this study, the action set of the agent, denoted by A_t, is in one-to-one correspondence with the neighborhood-operator set Ω, namely,

$$A_t = \Omega. \tag{39}$$

Each action $\omega_t \in A_t$ represents a local perturbation or structural optimization operation applied to the current TD-TOP solution. At each iteration of the algorithm, the agent selects an operator ω from the set of operators and applies it to the current solution to generate a new candidate solution. Accordingly, the policy function can be written as

$$\pi_\theta(\omega_t \mid s_t, h_t) = \Pr\{A_t = \omega_t \mid S_t = s_t, H_t = h_t, \theta_t = \theta\}, \tag{40}$$

where h_t denotes the hidden state of the LSTM, which captures the historical information and temporal dependencies of the search process,

and θ denotes the policy parameter. This design enables the agent to make decisions not only based on the characteristics of the current solution but also on the sequence of operator selections from previous steps, thereby achieving a balance between exploration and exploitation.

Algorithm 1 PPO–VNS with simulated annealing.

Require: cooling factor α, initial temperature T_0, max iterations $MaxIter$, trained policy network $\pi_\theta(\cdot)$
Ensure: best solution S^{best}

construct initial solution S_0
$S \leftarrow S_0$, $S^{best} \leftarrow S_0$
$T \leftarrow T_0$, $iter \leftarrow 0$
while $iter < MaxIter$ **do**
  extract state feature $x(S)$ from current solution
  select a neighborhood/operator $a \sim \pi_\theta(\cdot \mid x(S))$
  $S' \leftarrow a(S)$
  $\Delta \leftarrow f(S') - f(S)$
  if $\Delta > 0$ **then**
    $S \leftarrow S'$
  else
    accept S' with probability $\exp(\Delta/T)$
  end if
  if $f(S) > f(S^{best})$ **then**
    $S^{best} \leftarrow S$
  end if
  $T \leftarrow \alpha \cdot T$
  $iter \leftarrow iter + 1$
end while
return S^{best}

4.4.2. Reward function

In the PPO–VNS framework, the reward function R_t is designed to reflect, in real time, the improvement in solution quality after an operator is applied, while jointly accounting for energy consumption and time-related factors. The reward function is defined as follows:

$$R_t = \Delta I_t + \beta_1 \Delta W_t + \beta_2 \Delta C_t, \tag{41}$$

where the difference terms are defined as

$$\Delta I_t = I(S'_t) - I(S_t), \tag{42}$$

$$\Delta W_t = W(S_t) - W(S'_t), \tag{43}$$

$$\Delta C_t = C(S_t) - C(S'_t). \tag{44}$$

Here, $I(S)$ denotes the total collected information value of solution S, $W(S)$ denotes the total truck–drone waiting time, and $C(S)$ denotes the total UAV energy consumption. The parameters β_1 and β_2 are weighting coefficients used to balance the contributions of waiting-time reduction and energy-consumption reduction to the overall reward. The term ΔI_t reflects the improvement in the task objective after operator ω_t is executed. The term ΔW_t reflects the improvement in the efficiency of truck–drone cooperative scheduling, while ΔC_t encourages solutions with lower energy consumption. The values of β_1 and β_2 are reported in Appendix A.3.

The reward function provides an integrated description of the dynamic changes in solution quality and resource utilization, thereby guiding the agent to maximize the value of information collected during the optimization process.

4.4.3. Training method

The operator-selection policy is trained using Recurrent Proximal Policy Optimization (Recurrent PPO), a policy-gradient method for optimizing it in the formulated MDP. Unlike value-based methods, Recurrent PPO directly optimizes a parameterized stochastic policy, while using a value function as the critic. At each iteration, the observation is defined as

$$o_t = [s_t, \mathrm{onehot}(\omega_{t-1})],$$

where s_t denotes the extracted state features and ω_{t-1} denotes the operator selected in the previous iteration. The LSTM hidden state h_t allows the policy to retain historical search information and capture temporal dependencies along the operator-selection trajectory. This structure enables the agent to make operator-selection decisions by jointly considering the current solution state and the recent sequence of selected operators.

As shown in Fig. 4(a), the policy network adopts a recurrent actor–critic architecture. The input consists of the extracted state features s_t and the one-hot encoding of the previous operator ω_{t-1}. These inputs are first processed by two fully connected layers with 128 units each, which form a shared feature-extraction backbone for the policy and value networks. The extracted features are then passed to an LSTM layer with 256 hidden units to model temporal dependencies in the search trajectory. The LSTM output is fed into two heads. The policy head outputs a categorical distribution $\pi_\theta(\omega_t \mid o_t, h_t)$ over the neighborhood-operator set, which is used for operator selection. The value head outputs the state-value estimate $V_\phi(o_t, h_t)$, which is used for advantage estimation.

Fig. 4(b) illustrates the interaction flow between the policy network and the VNS search environment. At iteration t, state features are extracted from the current solution S_t and combined with the previous-operator encoding to form the observation o_t. The observation is fed into the recurrent policy network, which selects an operator ω_t according to the policy distribution $\pi_\theta(\omega_t \mid o_t, h_t)$. The selected operator is then executed within VNS to generate a candidate solution S'_t. The candidate solution is evaluated by the simulated annealing acceptance mechanism, which determines whether S'_t is accepted as the next solution S_{t+1}.

During offline training, the transition information generated by the current policy is collected as on-policy rollout data. The rollout sequence includes the observation, the recurrent hidden state, the selected operator, the reward, the next observation, the value estimate, and the policy log-probability. These data are stored in a sequence-aware on-policy rollout buffer and are used to compute the generalized advantage estimate (GAE). After each PPO update, the rollout buffer is cleared, and new rollout data are collected using the updated policy. During testing, the trained policy is embedded into VNS for online operator selection.

The policy is updated using the PPO clipped objective:

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\,\hat{A}_t, \mathrm{clip}\left(\rho_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right)\right], \tag{45}$$

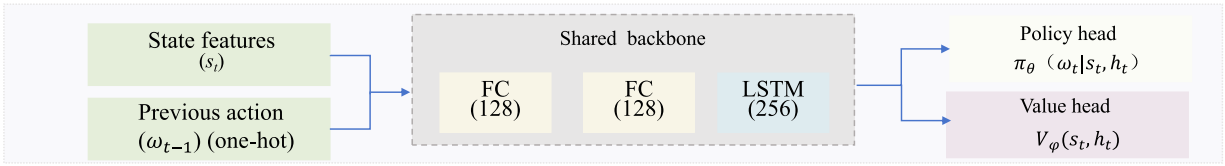
where

$$\rho_t(\theta) = \frac{\pi_\theta(\omega_t \mid o_t, h_t)}{\pi_{\theta_{\mathrm{old}}}(\omega_t \mid o_t, h_t)}. \tag{46}$$

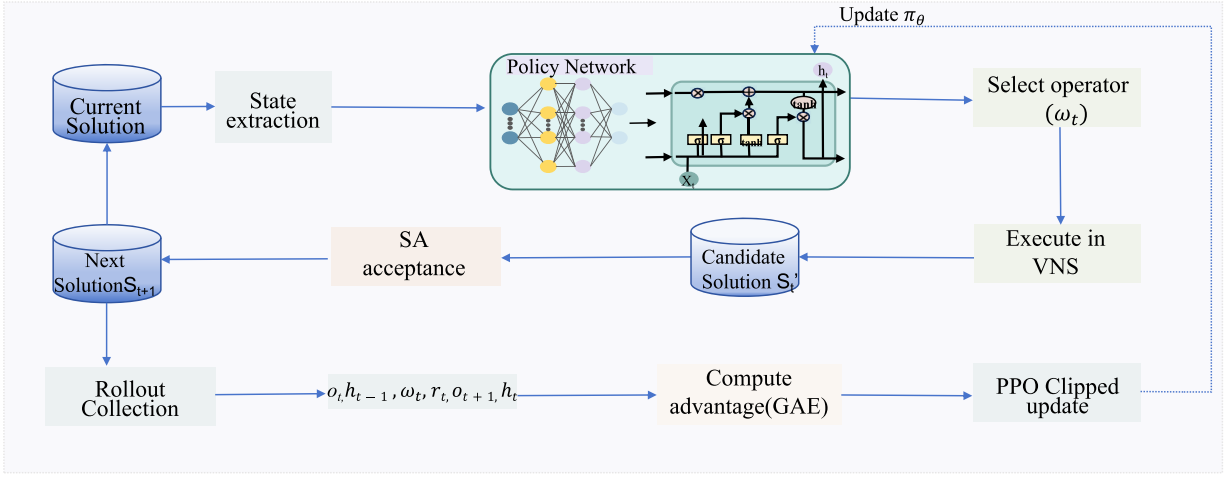
Here, $\rho_t(\theta)$ denotes the probability ratio between the new and old policies, and $\hat{A}_t$ denotes the generalized advantage estimate. The clipping mechanism constrains the policy update within a limited range, thereby improving training stability. For reproducibility, Appendix Algorithm A.1 provides the detailed rollout collection and recurrent PPO training procedure used in the PPO–VNS framework.

5. Numerical experiment

The performance of PPO–VNS is evaluated on benchmark instances of different sizes. For small-scale instances, the obtained solutions are compared with the optimal solutions produced by IBM ILOG CPLEX Optimization Studio 22.1.0. For large-scale instances, PPO–VNS is compared with VNS, Adaptive Variable Neighborhood Search (AVNS) (Madani et al., 2024), Variable Neighborhood Search combined with Simulated Annealing (VNS–SA) (Jiang et al., 2025), Adaptive Variable Neighborhood Search combined with Simulated Annealing (AVNS–SA), and DQN–VNS. All experiments are conducted on a computer equipped with an Intel(R) Core(TM) i9-13900K CPU at 3.0 GHz, an



(a) Policy Network Architecture



(b) Training and Search Interaction Flow

Fig. 4. Network and training flow of the PPO–VNS operator-selection policy.

NVIDIA GeForce RTX 4090D GPU, and 128 GB of RAM. All algorithms and test programs are implemented in Python 3.13.

5.1. Experimental design

5.1.1. Parameter settings

Based on related studies on truck–drone information collection tasks and the publicly available specifications of the DJI Air 3 series, we set the truck cruising speed to 50 km/h and assume no endurance limit for the truck within the planning horizon. We set the UAV cruise speed to 72 km/h and the battery capacity to 4276 mAh.

The training and tuning procedure is described below. The hyperparameters of PPO–VNS and DQN–VNS were tuned independently because the two algorithms have different learning mechanisms. PPO is an on-policy policy-gradient method based on rollout trajectories, whereas DQN is an off-policy value-based method using experience replay. Therefore, some hyperparameters, such as batch size and network size, do not play identical roles in the two algorithms and were not forced to take the same values.

For both PPO–VNS and DQN–VNS, preliminary tuning experiments were conducted on the same representative training datasets with 100–200 nodes. The Recurrent PPO policy is trained offline before computational testing and then embedded in the VNS framework for online operator selection. The policy is not retrained for each test instance. In this study, the same trained policy is used for all test instances, and the hyperparameters are held fixed throughout the experiments. Therefore, the reported results evaluate the learned operator-selection policy under a fixed deployment setting rather than instance-specific retraining. The CPU time reported in the computational tables refers to the online solving time of PPO–VNS after the PPO policy has been trained. The offline training time of the PPO policy is not included in the reported CPU time. Candidate configurations were evaluated according to convergence stability and solution quality. For each algorithm, the best-performing configuration was selected and held fixed across all testing instances to ensure a fair comparison. The reward-function weights β_1 and β_2 were also selected through preliminary tuning experiments and fixed throughout all experiments.

When simulated annealing is enabled, the temperature is decreased exponentially according to $\mathcal{T}_{k+1} = \alpha \mathcal{T}_k$, with $\mathcal{T}_0 = 100$, $\alpha = 0.99$, and $\mathcal{T}_{\min} = 0.01$. The overall search is terminated after 1000 iterations. The complete hyperparameter values of PPO–VNS and DQN–VNS, the reward-function weights, and the simulated-annealing parameters are reported in Appendix A.3.

5.1.2. Dataset settings

The training dataset consists of 1,000 randomly generated instances. Each instance contains one depot, $m_r \in [5, 15]$ road nodes, and $n_u \in [50, 200]$ candidate information-collection nodes distributed within a 25×25 km^2 region. Service times are sampled from a uniform distribution on $[0, 3]$ minutes, and fixed information values p_i are drawn from scenario-specific base distributions and truncated to $[1, 100]$. The spatial distribution of candidate nodes follows three patterns: R-type, C-type, and RC-type.

Two datasets are used for testing. The small-scale dataset, adapted from Morandi et al. (2024), consists of three groups with 10, 20, and 30 nodes, respectively. The original coordinates and node information values are retained, four road nodes are added to each instance, and the mission time limit is set to $T = 200$ min. The large-scale dataset, adapted from Uchoa et al. (2017), contains 100 instances, each with 100 to 1000 nodes. Each instance is rescaled to a 25×25 km^2 region and augmented with slightly perturbed road nodes to ensure feasible truck-supported UAV operations, with the mission time limit set to $T = 400$ min.

5.2. Experimental results

This section presents the computational results of PPO–VNS. Comparisons are conducted separately for small-scale and large-scale instances. For small-scale instances, PPO–VNS is compared with CPLEX

reference solutions. Specifically, CPLEX proves optimality for the 10- and 20-node instances, whereas for the 30-node instances, it reaches the one-hour time limit and returns incumbent-feasible solutions. For large-scale instances, PPO–VNS is benchmarked against five baselines: classical VNS, VNS–SA, AVNS, AVNS–SA, and DQN–VNS developed under the DRLH paradigm (Kallestad et al., 2023). In addition, sensitivity analyses are conducted to evaluate the robustness of the proposed method further. For each stochastic algorithm, all test instances are independently executed 10 times, and the average result is reported to reduce the influence of randomness. The results are evaluated in terms of solution quality and run-to-run stability.

5.2.1. Small-scale case analysis compared to CPLEX

To assess the performance of PPO–VNS on small-scale instances, comparative experiments are conducted against CPLEX. As shown in Table 3, for PPO–VNS we report both the best objective value over multiple runs, denoted by I_{best}, and the average objective value, denoted by I_{avg}. For the 10- and 20-node instances, the CPLEX solutions are proven optima. In these cases, PPO–VNS achieves almost the same objective values as CPLEX: I_{best} matches the CPLEX optimum in all cases, and the relative gap of I_{avg} with respect to CPLEX remains within 0.5%. CPLEX is faster than PPO–VNS on several 10- and 20-node instances because these instances are small enough that the exact solver can efficiently prove optimality, whereas PPO–VNS still follows a fixed iterative search process.

The relative gap to the CPLEX reference value is computed as

$$\text{GAP}(\%) = \frac{I_{\text{avg}} - I_{\text{CPLEX}}}{I_{\text{CPLEX}}} \times 100\%, \tag{47}$$

where I_{CPLEX} denotes the proven CPLEX optimum for the 10- and 20-node instances and the one-hour CPLEX incumbent for the 30-node instances. Therefore, the reported GAP is a relative performance gap with respect to the CPLEX reference value, rather than the CPLEX optimality gap. A positive GAP indicates that the average objective value obtained by PPO–VNS is higher than the corresponding CPLEX reference value, whereas a negative GAP indicates the opposite.

For the 30-node instances, CPLEX cannot prove optimality within the one-hour time limit and only returns incumbent feasible solutions. Therefore, the CPLEX results for these instances are treated as time-limited reference solutions rather than certified optima. In contrast, PPO–VNS obtains high-quality solutions within approximately 15–20 seconds. Notably, I_{best} is not lower than the one-hour CPLEX incumbent on all 30-node instances and exceeds this incumbent in several cases. This observation does not imply that PPO–VNS outperforms the exact optimum, since the CPLEX solutions for these instances are not certified optimal. Rather, it indicates that PPO–VNS can find competitive feasible solutions more efficiently under the imposed time limit. The average GAP of I_{avg} with respect to the CPLEX incumbent over the 30-node instances is about 0.75%, with values ranging from −3.60% to 6.77%. These results suggest that PPO–VNS can substantially reduce computation time while maintaining competitive solution quality, showing good computational efficiency and scalability.

5.2.2. Results on large-scale test instances

To evaluate the performance of the proposed method across different instance scales and spatial distributions, comparative experiments are conducted on benchmark instances with node sizes ranging from 100 to 1000. Six algorithms are compared: PPO–VNS, DQN–VNS, VNS, VNS–SA, AVNS, and AVNS–SA. According to the spatial distribution of nodes, the instances are grouped into three categories: R-type (random), C-type (clustered), and RC-type (mixed). The results for R-type, C-type, and RC-type instances are reported in Tables 4–6, respectively. Across all three categories, PPO–VNS achieves the highest average I_{avg}. In terms of the category-level average objective value, PPO–VNS obtains the highest average I_{avg} in all three categories, while DQN–VNS provides the most competitive performance among the benchmark methods. VNS and VNS–SA generally form an intermediate group, whereas AVNS and AVNS–SA obtain lower average objective values in most categories. At the instance level, the performance gap between PPO–VNS and DQN–VNS is relatively small in several cases, and some heuristic baselines can still produce competitive solutions for specific instance structures. Relative to DQN–VNS, PPO–VNS improves the average objective value by about 1.2% on R-type instances, 1.9% on C-type instances, and 0.9% on RC-type instances. Compared with VNS, the corresponding gains are about 9.1%, 5.5%, and 11.2%, while the improvements over VNS–SA are 11.1%, 7.1%, and 11.8%. In addition, PPO–VNS outperforms AVNS and AVNS–SA by approximately 14.4%–19.5%. These results suggest that the PPO-based neighborhood-selection strategy provides a stable advantage in average solution quality across different spatial distributions. In particular, PPO–VNS achieves the highest category-level average I_{avg} in all three instance groups, while DQN–VNS remains the closest competing method. This indicates that recurrent policy-guided operator selection can improve the effectiveness of VNS search, although the magnitude of improvement is instance- dependent.

In terms of running time, DQN–VNS is the fastest on average, followed by PPO–VNS. The two RL-guided methods generally require less online search time than the four VNS-based heuristic baselines, especially on larger instances. AVNS and AVNS–SA show intermediate runtime, whereas VNS and VNS–SA are generally the slowest. As the instance size increases, the runtime of all methods grows, but the increase for the RL-guided methods is considerably smaller. In the largest instances, the runtime of PPO–VNS and DQN–VNS is only about 30%–40% of that required by classical VNS.

Fig. 5 presents the task coverage ratio for all test instances. For the relatively small instances in the large-scale test set, all methods achieve high coverage levels. As the instance size increases and resource constraints become tighter, the coverage ratio generally declines. PPO–VNS attains the highest or nearly highest coverage in most instances. DQN–VNS also remains highly competitive and usually stays close to the leading group. In contrast, AVNS and AVNS–SA, together with VNS–SA in some cases, fall more frequently into the lower-coverage range on larger instances. Although all VNS-based heuristics exhibit a similar downward trend, the figure indicates that refining the acceptance criterion or introducing adaptive tuning alone does not always lead to stable performance improvements under tighter constraints.

Table 3
Comparison between CPLEX and PPO–VNS on benchmark instances.

Instance	CPLEX		PPO–VNS			GAP
	I_{CPLEX}	CPU (s)	I_{best}	I_{avg}	CPU (s)	
poi-10-1	**411**	0.56	**411**	**411**	15.63	–
poi-10-2	**430**	0.22	**430**	**430**	15.81	–
poi-10-3	**449**	0.22	**449**	**449**	15.13	–
poi-10-4	**468**	0.41	**468**	**468**	15.76	–
poi-10-5	**387**	0.483	**387**	**387**	15.32	–
poi-20-1	**941**	11.55	**941**	**941**	15.48	–
poi-20-2	**970**	16.44	**970**	968	16.69	−0.21%
poi-20-3	**996**	35.28	**996**	991	16.18	−0.50%
poi-20-4	**1021**	11.39	**1021**	**1021**	15.23	–
poi-20-5	**957**	2.20	**957**	**957**	16.32	–
poi-30-1	**1334**	3619.44	**1334**	1286	17.15	−3.60%
poi-30-2	1319	3601.22	**1383**	1320	15.49	0.08%
poi-30-3	1300	3603.50	**1410**	1388	16.76	6.77%
poi-30-4	1280	3603.34	**1344**	1306	19.84	2.03%
poi-30-5	**1452**	3603.66	**1452**	1430	17.95	−1.52%

Note: I_{CPLEX} denotes the total collected information value reported by CPLEX. For 10- and 20-node instances, the CPLEX values are proven optima. For 30-node instances, CPLEX reaches the one-hour time limit, and the reported values are time-limited incumbent solutions rather than certified optima. GAP denotes the relative performance gap between I_{avg} and the corresponding CPLEX reference value, not the CPLEX optimality gap.

Table 4
Comparison results of six algorithms on R-type instances.

Instance	Cust	PPO–VNS			DQN–VNS			VNS			VNS–SA			AVNS			AVNS–SA		
		CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}
X-n110-k13	R	33.13	**7686**	7478	27.25	7655	7441	48.61	7527	7342	45.69	7308	7123	49.21	7308	7121	47.19	7308	6823
X-n115-k10	R	34.03	8055	7765	26.93	7979	7761	50.86	7979	7785	48.69	7915	7728	34.56	**8164**	7976	33.47	7798	7291
X-n139-k10	R	38.49	**10227**	9931	34.76	10,146	9929	75.16	10,143	9951	74.35	10,065	9879	74.60	9235	9053	44.00	9382	8817
X-n143-k7	R	37.67	**9228**	8751	29.78	8963	8758	66.29	8390	8197	55.75	8571	8385	45.18	8160	7988	37.53	8154	7679
X-n167-k10	R	49.32	11,782	11,592	47.99	**11808**	11,582	118.34	11,688	11,499	103.18	11,622	11,436	99.43	11,134	10,929	53.78	11,394	10,909
X-n176-k26	R	59.84	**12846**	12,540	71.25	12,756	12,537	163.98	12,403	12,212	165.09	12,403	12,217	103.61	12,330	12,145	80.67	12,460	11,886
X-n186-k15	R	80.74	**13972**	13,757	85.59	**13972**	13,755	231.03	**13972**	13,786	224.20	**13972**	13,789	165.53	**13972**	13,790	110.76	**13972**	13,410
X-n209-k16	R	89.27	**15634**	15,352	111.83	15,565	15,348	279.89	15,582	15,396	282.24	15,458	15,271	248.95	15,392	15,187	143.13	15,458	14,893
X-n219-k71	R	97.47	**16380**	16,163	125.86	16,314	16,089	278.53	16,240	16,012	278.32	16,164	15,938	211.35	16,083	15,897	138.32	16,083	15,621
X-n237-k14	R	98.43	**17610**	17,327	113.46	17,396	17,107	291.84	17,187	16,654	289.77	17,052	16,497	238.96	16,734	16,386	143.01	17,302	16,748
X-n242-k50	R	95.66	**18102**	17,743	103.34	18,068	17,732	289.44	17,916	17,283	289.24	17,200	16,512	131.56	16,956	16,521	108.50	17,148	16,493
X-n261-k13	R	100.59	**19574**	19,153	104.54	**19574**	19,183	309.73	19,391	18,579	308.62	19,266	18,429	171.93	18,852	17,987	148.01	18,912	17,857
X-n280-k17	R	101.11	**19907**	19,411	113.15	19,765	19,287	312.86	17,512	16,897	314.62	17,195	16,531	201.05	15,761	14,196	114.12	16,598	15,132
X-n294-k50	R	112.92	**21363**	20,795	113.91	19,990	19,396	327.78	18,501	17,736	328.44	18,507	17,674	153.31	17,705	16,042	146.59	17,946	16,311
X-n298-k31	R	97.84	**21292**	20,658	111.66	20,885	20,221	334.57	18,895	18,052	332.84	18,468	17,596	199.15	18,052	15,998	202.10	17,887	15,924
X-n322-k28	R	123.81	**22068**	21,366	115.89	20,974	20,281	337.53	20,591	19,648	338.22	20,405	19,438	131.24	18,433	16,299	132.16	18,588	16,713
X-n331-k15	R	114.55	**22947**	22,172	121.41	21,607	20,834	359.72	19,747	18,829	342.42	19,260	18,317	158.00	18,113	15,688	144.24	18,016	15,891
X-n336-k84	R	109.21	20,676	19,848	119.73	**21541**	20,669	336.52	18,561	17,695	342.29	18,151	17,286	133.39	19,456	17,031	126.33	17,289	15,402
X-n376-k94	R	126.54	**23252**	22,371	135.95	23,167	22,302	362.44	20,612	19,587	361.79	20,201	19,165	129.69	20,867	17,922	122.52	20,699	18,034
X-n384-k52	R	128.03	**22025**	21,106	133.64	21,946	21,052	374.20	18,715	17,768	368.17	17,092	16,139	187.88	17,096	14,851	147.68	18,142	15,809
X-n429-k130	R	122.92	**22213**	21,260	155.79	21,775	20,809	427.50	19,274	18,241	423.76	18,448	17,423	193.44	17,913	15,549	151.48	19,539	16,804
X-n449-k29	R	127.64	**22905**	21,929	139.58	22,188	21,232	406.14	20,143	19,096	407.50	19,776	18,711	188.30	18,778	16,053	167.50	18,970	16,406
X-n469-k138	R	132.43	22,160	21,168	145.42	**22808**	21,855	418.24	19,772	18,729	419.13	19,505	18,489	198.11	19,818	16,903	188.22	19,746	17,019
X-n524-k53	R	160.10	**22331**	21,706	141.83	**22331**	21,758	472.79	20,536	20,089	460.91	19,233	18,819	273.09	18,484	17,692	242.05	18,102	17,329
X-n548-k56	R	178.10	**21252**	20,689	147.78	20,547	19,903	506.52	19,587	18,892	505.28	18,406	17,853	228.95	19,398	18,513	208.58	19,588	18,105
X-n593-k92	R	200.75	**21874**	21,211	226.83	20,678	20,012	611.08	20,212	19,437	608.45	20,112	19,357	286.25	20,028	18,842	243.30	20,207	18,563
X-n619-k62	R	213.54	**24421**	23,845	243.69	23,801	23,169	677.53	21,568	20,815	679.23	21,660	20,902	373.15	20,996	19,321	297.86	21,368	19,601
X-n670-k130	R	239.01	**21556**	20,972	208.34	21,467	20,821	604.14	19,390	18,646	603.60	19,487	18,792	504.92	17,361	16,578	502.32	17,361	16,089
X-n736-k15	R	232.79	**21900**	21,275	243.42	21,616	20,970	700.08	19,796	19,051	702.89	19,700	18,946	515.45	19,709	18,124	345.80	20,091	18,347
X-n786-k71	R	261.99	**21259**	20,614	224.14	20,876	20,234	685.95	20,094	19,329	682.56	19,125	18,410	313.73	20,681	18,907	336.66	19,314	17,698
X-n801-k40	R	260.98	**21410**	20,755	248.45	21,122	20,480	714.49	20,953	20,188	729.67	20,951	20,218	421.89	20,246	18,619	351.30	19,952	18,274
X-n895-k37	R	291.76	**22693**	22,038	267.40	22,511	21,867	808.50	20,558	19,793	816.33	20,655	19,891	459.07	21,664	19,903	396.73	21,454	19,728
X-n936-k151	R	295.79	20,961	20,317	291.45	**21059**	20,415	781.54	20,082	19,317	777.48	19,199	18,524	428.63	19,004	17,459	338.99	20,373	18,796
X-n1001-k53	R	292.47	22,051	21,398	245.28	**22734**	22,091	759.63	20,778	20,013	767.10	20,975	20,229	484.84	20,678	18,936	492.71	20,776	19,041
AVERAGE				**18366**			18,144			16,840			16,527			15,600			15,572

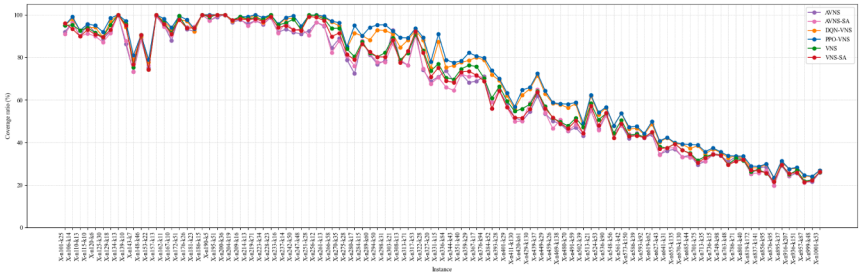


Fig. 5. UAV coverage ratio of six algorithms across all instances.

Three representative instances, namely X-n181-k23, X-n573-k30, and X-n876-k59, were selected to compare the objective-value distributions of the six algorithms under different problem scales, with 30 repeated runs conducted for each instance. As shown in Fig. 6, for the smaller instance X-n181-k23, the boxplots of all six algorithms are relatively compact, with small interquartile ranges. This indicates that, under relatively relaxed resource constraints, all algorithms can stably obtain solutions of similar quality. PPO–VNS has a slightly higher median, while DQN–VNS remains very close and also belongs to the leading group. As the problem scale increases, the differences among algorithms become more pronounced in X-n573-k30. PPO–VNS achieves the highest median among the methods, with a relatively concentrated distribution, demonstrating strong solution quality and good robustness. DQN–VNS forms the second-best group and exhibits a comparable level of stability, whereas VNS and VNS–SA remain in the intermediate range. AVNS shows the largest variability, and AVNS–SA also exhibits a noticeably wider spread than PPO–VNS and DQN–VNS. For the largest instance, X-n876-k59, PPO–VNS and DQN–VNS remain the best-performing methods, with close medians and relatively small dispersions. In contrast, the other heuristic variants show lower medians and/or greater variability. Overall, these results indicate that learning-guided methods can obtain high-quality solutions more consistently on more challenging instances. However, the performance gap between PPO–VNS and DQN–VNS remains instance-dependent.

5.2.3. *Sensitivity analysis*

To quantitatively evaluate the impact of UAV fleet size, we selected nine representative instances spanning 200–1000 nodes in 100-node intervals and varied the number of UAVs from two to six. The results are reported in Fig. 7. The thin curves represent the results of individual instances, while the thick curve reports the mean objective value with 95% confidence intervals across the tested instances. Overall, increas-

Table 5
Comparison results of six algorithms on C-type instances.

Instance	Cust	PPO–VNS			DQN–VNS			VNS			VNS–SA			AVNS			AVNS–SA		
		CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}
X-n106-k14	C (3)	32.54	**7704**	7529	28.88	7646	7481	64.09	7448	7252	53.24	7306	7108	47.15	7501	7310	34.73	7507	7145
X-n125-k30	C (5)	35.90	**8526**	8311	29.86	8303	8089	49.14	8345	8142	44.42	8345	8139	54.16	8259	8051	41.95	8116	7698
X-n134-k13	C (4)	48.75	**10068**	9852	38.92	**10068**	9861	101.85	**10068**	9861	101.50	**10068**	9866	66.11	**10068**	9855	62.25	**10068**	9873
X-n153-k22	C (3)	38.56	**9347**	9135	37.12	8681	8473	88.37	8802	8603	82.62	8797	8602	68.76	8876	8689	53.48	9003	8546
X-n157-k13	C (3)	66.69	**11825**	11,607	70.10	**11825**	11,602	159.56	**11825**	11,620	167.76	**11825**	11,616	140.83	11,768	11,552	86.41	11,768	11,439
X-n181-k23	C (6)	63.75	**12988**	12,772	75.90	12,507	12,295	171.75	12,810	12,609	172.87	12,784	12,591	133.24	12,617	12,423	95.89	12,842	12,369
X-n190-k5	C (3)	78.92	**14151**	13,936	93.28	**14151**	13,946	242.16	14,046	13,852	239.43	14,097	13,893	197.78	13,871	13,666	138.76	13,881	13,500
X-n200-k36	C (8)	102.80	**14957**	14,741	109.15	**14957**	14,739	277.12	**14957**	14,749	264.32	**14957**	14,756	229.78	**14957**	14,738	157.99	**14957**	14,731
X-n214-k13	C (4)	119.34	**15996**	15,781	112.73	15,955	15,728	254.23	15,898	15,661	232.95	15,835	15,606	160.41	15,585	15,299	157.32	15,466	14,880
X-n228-k23	C (8)	97.42	**17185**	16,896	90.70	17,185	16,887	300.29	17,185	16,876	305.34	17,012	16,698	180.22	16,922	16,507	154.88	16,922	16,199
X-n247-k48	C (4)	92.84	**17776**	17,424	120.74	17,356	16,981	287.96	17,489	17,113	284.01	17,428	17,046	108.09	17,281	16,649	107.28	17,566	16,619
X-n256-k12	C (8)	87.14	**19314**	18,897	128.11	**19314**	18,869	323.06	19,248	18,790	329.29	19,128	18,663	190.99	18,787	17,890	195.07	18,787	17,605
X-n275-k28	C (3)	89.75	**18065**	17,574	123.05	17,822	17,297	310.52	17,909	17,386	312.79	17,336	16,798	175.12	16,771	15,628	157.13	17,191	15,756
X-n284-k15	C (8)	91.02	**19629**	19,066	119.63	19,305	18,719	328.42	19,210	18,609	323.60	19,021	18,402	153.18	19,249	17,821	125.89	19,256	17,558
X-n303-k21	C (8)	102.94	**21471**	20,823	126.65	21,425	20,732	340.30	20,741	20,052	341.15	20,547	19,850	144.99	21,443	19,687	161.67	20,276	18,322
X-n317-k53	C (4)	114.58	22,372	21,650	114.33	**22425**	21,670	381.57	21,939	21,175	391.67	22,167	21,384	150.53	22,209	20,190	147.21	21,895	19,719
X-n351-k40	C (3)	94.60	21,520	20,715	133.04	**21536**	20,691	352.84	20,543	19,696	351.23	20,191	19,335	209.26	19,971	17,687	167.99	19,908	17,519
X-n367-k17	C(4)	117.18	22,457	21,544	145.92	**22710**	21,726	406.68	21,713	20,788	391.88	20,529	19,595	225.78	19,668	17,211	196.78	20,454	17,913
X-n401-k29	C (6)	101.16	**20072**	19,517	135.40	18,898	18,453	350.12	18,527	18,012	348.31	17,410	16,897	147.26	18,083	15,492	162.60	17,414	14,777
X-n411-k130	C (5)	102.11	**19499**	18,956	137.28	19,438	18,892	361.96	18,156	17,693	361.90	17,101	16,548	141.59	17,096	14,423	216.18	16,362	13,650
X-n459-k26	C (4)	129.76	**21808**	21,092	137.18	20,768	20,011	407.15	19,079	18,307	396.05	19,163	18,401	172.06	18,465	15,717	171.52	17,020	14,235
X-n480-k70	C (8)	124.55	**21441**	20,688	153.85	20,762	19,987	442.67	17,842	17,189	434.47	17,299	16,632	240.73	16,733	13,918	209.19	16,828	13,985
X-n502-k39	C (3)	121.52	**19621**	18,985	137.48	18,197	17,659	432.48	18,733	18,041	438.42	17,353	16,679	265.02	16,794	13,905	197.24	17,261	14,365
X-n536-k90	C (7)	144.13	**23911**	23,079	188.94	22,948	22,203	622.39	22,493	21,675	624.84	22,664	21,889	294.19	22,124	19,171	277.70	22,035	19,071
X-n573-k150	C (3)	163.24	**21734**	21,016	210.97	21,285	20,578	620.36	19,701	19,026	654.83	19,520	18,856	431.50	18,947	18,012	289.76	19,988	19,234
X-n627-k43	C (5)	235.67	**20754**	20,011	201.52	20,747	20,009	639.34	19,165	18,409	658.24	18,598	17,924	344.08	16,956	16,143	319.25	16,865	16,198
X-n655-k135	C (4)	191.65	**21227**	20,473	274.22	21,025	20,312	804.13	20,648	19,914	826.14	20,556	19,821	383.22	18,993	18,179	365.70	19,385	18,542
X-n713-k35	C (3)	239.53	**20972**	20,228	247.13	20,611	19,945	786.02	19,458	18,722	777.44	18,969	18,253	689.10	17,898	17,035	689.39	17,898	17,129
X-n749-k98	C (8)	258.56	**21052**	20,305	271.73	20,955	20,218	818.06	20,183	19,436	820.73	20,084	19,318	410.08	19,986	19,158	355.54	20,277	19,503
X-n819-k172	C (6)	257.55	**19660**	18,904	217.50	18,909	18,264	694.86	17,458	16,813	680.23	18,135	17,461	371.79	16,772	15,998	362.58	16,770	15,987
X-n876-k95	C (5)	227.44	**17467**	16,782	216.29	16,601	15,988	636.31	14,952	14,289	643.98	15,050	14,395	561.14	13,484	12,716	559.52	13,484	12,761
X-n999-k48	C (6)	267.62	**18941**	18,198	255.88	18,749	18,027	725.77	17,568	16,904	733.78	17,273	16,598	479.74	16,676	15,891	396.74	17,272	16,518
AVERAGE				**17390**			17,073			16,477			16,238			15,207			15,092

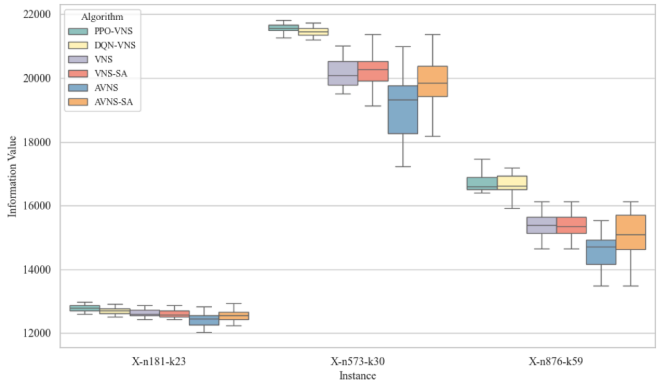


Fig. 6. Objective-value distributions of six algorithms on instances of different sizes.

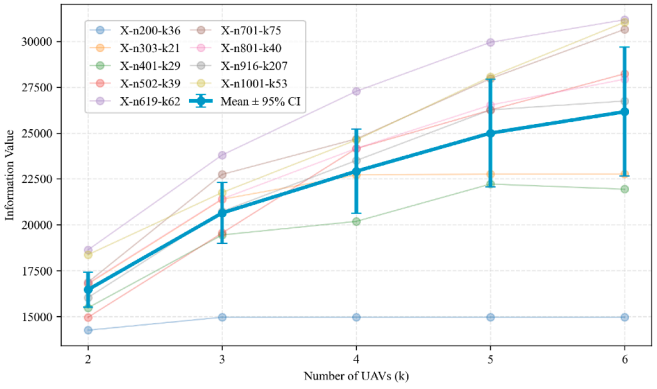


Fig. 7. Effect of the number of UAVs on the total collected information value across different instances.

ing the UAV fleet size improves the value of the collected information, indicating that additional UAVs enhance the task-coverage capability of the truck–drone system.

To further support the observed trend, we compute the marginal gain between adjacent parameter levels. For a parameter setting x_j, the marginal gain is defined as

$$\mathrm{MG}(x_j) = \frac{\bar{I}(x_j) - \bar{I}(x_{j-1})}{\bar{I}(x_{j-1})} \times 100\%, \tag{48}$$

where $\bar{I}(x_j)$ denotes the mean collected information value under parameter setting x_j. As shown in Fig. 7, the mean objective value increases rapidly when the fleet size is small, whereas the incremental improvement becomes smaller at larger fleet sizes. Specifically, the marginal gain decreases from 25.43% when the fleet size increases from two to three UAVs to 11.03% when it increases from three to four UAVs. The marginal gain further decreases to 9.08% from four to five UAVs and 4.66% from five to six UAVs. This decreasing pattern provides quantitative support for the diminishing marginal-gain effect. At the instance level, smaller-scale cases show only limited improvement beyond three to four UAVs, whereas larger-scale cases retain a more pronounced upward trend. Nevertheless, additional gains beyond five UAVs are relatively limited under the same time and energy constraints. The corresponding marginal-gain values are reported in Appendix A.4.

We further conducted a sensitivity analysis to evaluate the impact of UAV battery capacity E on objective performance. Experiments were performed on nine instances near the 300-node scale with $E \in \{1900, 2850, 4276, 5100\}$ mAh. These levels follow the specifications of mainstream multi-rotor UAVs and cover a typical practical capacity

Table 6
Comparison results of six algorithms on RC-type instances.

Instance	Cust	PPO–VNS			DQN–VNS			VNS			VNS–SA			AVNS			AVNS–SA		
		CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}	CPU (s)	I_{best}	I_{avg}
X-n101-k25	RC (7)	29.24	**7262**	7062	24.04	7200	7008	46.23	7143	6957	39.85	7181	6998	39.89	6895	6721	31.21	6820	6358
X-n120-k6	RC (8)	36.07	**8489**	8299	31.61	8301	8094	61.98	8156	7951	55.03	8084	7892	48.72	8159	7984	38.82	8008	7521
X-n129-k18	RC (8)	38.00	**9376**	9166	34.99	9292	9104	75.49	9157	8962	75.27	8897	8705	53.99	8819	8643	42.71	8760	8294
X-n148-k46	RC (7)	39.00	**10038**	9853	34.43	9881	9666	72.96	10,038	9833	84.57	10,038	9843	79.28	9778	9599	61.07	9778	9312
X-n162-k11	RC (8)	62.78	11,823	11,618	60.10	**12109**	11,913	161.17	11,731	11,536	153.20	11,689	11,497	118.93	11,609	11,432	58.28	11,532	11,076
X-n172-k51	RC (5)	68.92	**12893**	12,698	69.28	**12893**	12,690	166.08	12,836	12,641	165.51	12,583	12,391	155.15	12,796	12,619	139.85	12,729	12,253
X-n195-k51	RC (5)	88.88	**14534**	14,319	116.46	**14534**	14,349	257.78	**14534**	14,340	260.10	**14534**	14,342	186.00	14,415	14,238	184.68	14,460	13,988
X-n204-k19	RC (6)	84.47	**15124**	14,944	84.55	15,016	14,805	228.11	15,022	14,817	226.01	15,022	14,819	149.19	14,912	14,737	121.92	14,970	14,497
X-n223-k34	RC (5)	89.05	**16628**	16,421	94.76	16,524	16,311	290.22	16,360	16,198	285.81	16,232	16,059	118.61	16,075	15,901	139.44	16,129	15,683
X-n233-k16	RC (7)	90.65	**16871**	16,608	104.46	16,871	16,595	250.03	16,734	16,157	221.69	16,585	15,972	125.74	16,249	15,893	130.69	16,182	15,597
X-n251-k28	RC (3)	85.69	**18834**	18,515	99.55	**18834**	18,505	303.11	**18834**	18,172	302.87	18,729	17,984	192.55	17,767	17,284	244.10	17,454	16,819
X-n266-k58	RC (6)	101.03	**19781**	19,389	96.63	19,632	19,244	313.86	19,023	18,259	313.19	18,512	17,658	162.11	17,551	16,926	169.55	17,143	16,298
X-n270-k35	RC (5)	111.89	**19790**	19,343	107.41	19,539	19,088	306.06	19,300	17,984	305.60	18,966	17,893	128.07	18,532	17,805	103.71	18,385	17,302
X-n289-k60	RC (7)	98.20	20,208	19,687	120.85	**20829**	20,312	332.85	18,387	17,213	340.90	18,552	17,468	183.54	18,244	17,319	162.24	18,311	16,974
X-n308-k13	RC (6)	106.84	**21031**	20,443	113.11	20,934	20,342	321.70	18,805	17,768	321.03	18,470	16,987	206.21	18,651	16,987	142.44	18,813	15,947
X-n313-k71	RC (3)	108.42	**21411**	20,746	134.33	21,346	20,682	327.94	19,823	18,791	303.68	20,075	19,012	164.33	18,384	15,992	164.63	18,384	15,809
X-n327-k20	RC (7)	97.34	19,284	18,571	121.23	**19606**	18,878	300.02	18,822	17,735	299.65	18,119	17,031	129.94	17,517	15,342	132.50	17,193	14,651
X-n344-k43	RC (7)	90.74	**20547**	19,788	125.30	20,445	19,662	325.31	18,754	17,589	321.93	18,344	17,198	143.14	18,507	13,989	131.69	17,264	14,389
X-n359-k29	RC (7)	117.39	22,100	21,268	134.26	**22894**	22,045	377.25	21,412	20,286	385.79	20,633	19,486	202.56	19,027	16,215	164.56	19,903	17,128
X-n393-k28	RC (5)	129.55	**21596**	20,720	133.06	21,173	20,277	372.16	20,407	19,194	378.02	19,732	18,564	169.59	19,709	13,476	161.62	19,799	15,846
X-n420-k61	RC(3)	120.71	21,513	20,599	131.33	**22568**	21,631	367.51	18,935	17,742	368.64	17,396	16,189	180.00	16,846	12,018	166.53	16,844	12,919
X-n439-k37	RC (8)	148.52	**24892**	23,905	147.07	23,205	22,224	478.30	21,898	20,733	478.42	21,735	20,491	227.41	21,025	18,193	182.49	22,182	19,257
X-n491-k59	RC (6)	125.06	**22575**	22,012	147.50	22,335	21,758	426.94	19,712	19,137	426.01	19,162	18,703	192.78	17,672	16,895	187.72	18,328	17,683
X-n513-k21	RC (4)	146.79	**25360**	24,689	167.25	24,717	24,109	511.60	23,355	22,132	515.48	22,828	21,693	264.45	21,710	20,968	343.37	21,801	20,647
X-n561-k42	RC (7)	185.03	23,831	23,107	204.09	**24453**	23,867	607.31	22,196	20,842	614.91	21,253	20,519	563.19	20,877	19,543	560.20	20,877	19,721
X-n586-k39	RC (4)	198.40	**21744**	21,109	205.98	21,548	20,951	634.82	20,248	19,015	631.44	19,779	19,042	293.07	20,432	18,987	254.43	20,054	18,819
X-n641-k31	RC (8)	237.23	**21558**	20,915	228.27	21,375	20,782	664.57	18,633	17,208	661.40	18,726	18,011	497.10	17,868	16,921	282.86	18,737	17,562
X-n685-k44	RC (6)	232.60	**21814**	21,182	238.20	20,517	19,923	673.02	19,175	17,793	674.44	18,978	18,209	320.66	18,508	17,132	351.42	17,924	16,847
X-n701-k75	RC (7)	232.36	**22747**	22,001	210.09	22,383	21,791	600.81	17,973	16,498	598.50	17,301	16,647	354.00	16,625	15,788	292.31	17,495	16,238
X-n763-k48	RC (7)	260.56	**21173**	20,528	226.90	19,837	19,242	607.08	18,208	16,581	610.13	18,014	17,358	290.93	19,168	17,641	318.55	18,401	16,989
X-n837-k141	RC (7)	248.76	**19621**	19,003	223.84	19,235	18,641	653.92	18,465	16,923	646.99	17,881	17,136	296.65	19,142	17,529	336.36	17,194	15,741
X-n856-k95	RC (3)	254.91	**21121**	20,476	233.28	20,838	20,245	625.65	17,455	15,982	623.48	17,656	16,932	307.77	19,975	18,304	318.37	18,528	17,015
X-n916-k207	RC (6)	265.45	**20727**	20,054	243.23	19,953	19,360	648.06	18,482	16,895	650.14	18,779	18,027	359.51	17,694	16,019	318.01	17,992	16,439
X-n957-k87	RC (4)	259.28	**18987**	18,351	220.89	18,310	17,718	649.14	16,747	15,214	651.54	16,256	15,589	422.21	16,256	14,823	400.76	16,059	14,827
AVERAGE				**18159**			17,994			16,326			16,245			15,290			15,190

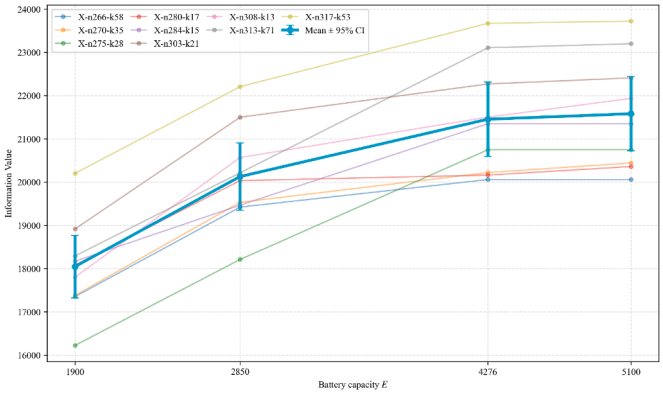


Fig. 8. Effect of UAV battery capacity on the total collected information value across different instances.

range. The results are shown in Fig. 8, where the error bars represent 95% confidence intervals across the tested instances.

The objective value shows an overall increasing tendency with E, indicating that a larger battery capacity can improve solution quality when UAV energy is restrictive. However, the improvement becomes much smaller in the high-capacity regime. Specifically, increasing the battery capacity from 1900 to 2850 mAh improves the mean objective value by 11.54%, while the marginal gain decreases to 6.59% when the capacity increases from 2850 to 4276 mAh. When the capacity further increases from 4276 to 5100 mAh, the marginal gain is only 0.59%. These results quantitatively support the observation that the benefit of increasing battery capacity gradually saturates. Once the energy budget is sufficiently relaxed, further increases in E provide limited additional improvement because the remaining bottlenecks are more likely to be related to mission duration, truck–drone synchronization, and the spatial distribution of candidate nodes. The corresponding marginal-gain values are reported in Appendix A.4.

6. Conclusion

This study formulated an information-collection TD-TOP in which one truck coordinates multiple UAVs under routing, synchronization, energy, and duration constraints. The proposed model explicitly captures multi-node UAV sorties, continuous truck progression, launch–recovery operations, bounded mutual waiting times, and a unified energy budget covering flight, service, and waiting activities. The problem was first formulated as an MILP model to characterize its operational structure and feasibility conditions. To solve large instances, we developed PPO–VNS by integrating Proximal Policy Optimization with variable neighborhood search. The framework combines adaptive clustering-based initialization, problem-specific truck–drone coordination neighborhoods, and a PPO-based operator-selection policy to improve search effectiveness while maintaining feasibility.

Computational results demonstrated that PPO–VNS consistently outperforms the benchmark methods in objective value across a wide range of instance sizes, while also showing stronger robustness in identifying feasible solutions under tight constraints. These results indicate that the proposed framework provides an effective and scalable approach for solving large-scale TD-TOP instances arising in practical information collection settings.

Future work will extend the current single-truck multi-UAV scheme to multi-truck multi-UAV systems, incorporate heterogeneous UAV platforms and additional operational constraints such as time windows, and move beyond operator selection toward more integrated learning of neighborhood selection and search-depth control.

CRediT authorship contribution statement

Yidan Wang: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Investigation; **Zhihao Luo:** Formal analysis, Data curation, Writing – review & editing, Supervision, Project administration; **Guohua Wu:** Methodology, Supervision, Writing – review & editing, Validation, Visualization; **Jianmai Shi:** Conceptualization, Supervision, Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A.

A.1. Complete feature components of the state representation

This appendix reports the complete feature components used by the Recurrent PPO agent. The observation consists of 20 scalar features and a one-hot encoding of the previously selected neighborhood operator. These features provide a compact, fixed-dimensional summary of the current search status, covering solution quality, feasibility pressure, search dynamics, and operator usage history. Instead of encoding the complete truck–drone routing solution, the state representation uses aggregate indicators to support adaptive neighborhood-operator selection.

The complete list of state-feature components is provided in Table A.1.

Table A.1
Complete feature list of the state representation.

Notation	Description
$I(S_t)$	Total collected information value of the current solution S_t.
$W^G(S_t)$	Total truck waiting time of the current solution.
$D^G(S_t)$	Total truck travel distance of the current solution.
$\tau^T(S_t)$	Remaining slack with respect to the maximum mission-duration constraint.
$C^U(S_t)$	Total UAV energy consumption of the current solution.
T_t^{SA}	Current temperature in the simulated-annealing process.
b_t^{acc}	Binary indicator of whether the newly generated solution is accepted.
$\mathrm{sgn}(R_{t-1})$	Sign of the reward obtained in the previous step.
n_t^{no}	Number of consecutive moves without improvement.
$F(S_t)$	Current evaluation score of the solution.
F_t^{best}	Best historical evaluation score found so far.
$F(S_{t-1})$	Evaluation score of the solution in the previous step.
$N^{\mathrm{vis}}(S_t)$	Number of UAV-served nodes in the current solution.
$\rho^I(S_t)$	Ratio of collected information value to the total available information value.
$\bar{L}^U(S_t)$	Average length of UAV missions.
$\tau_{\max}^{\mathrm{srv}}(S_t)$	Maximum service time among all UAV missions.
$\bar{\tau}^{\mathrm{srv}}(S_t)$	Average service time over all UAV missions.
ρ_t^{prog}	Ratio of the current search progress to the maximum search budget.
ℓ_t^{streak}	Consecutive usage length when the current operator is identical to the previous one.
$n_{\omega_{t-1}}^{\mathrm{use}}$	Cumulative usage count of the previous operator in the current search round.
$\mathrm{onehot}(\omega_{t-1})$	One-hot encoding of the neighborhood operator used in the previous step.

A.2. Implementation details of rollout processing and recurrent training

The main text presents the recurrent actor–critic network architecture and the PPO clipped objective. For reproducibility, this appendix summarizes the rollout collection and recurrent PPO training procedure used in the PPO–VNS framework. The procedure follows an on-policy training scheme: rollout data are collected using the current policy, used for PPO updates, and then cleared before the next rollout.

During training, rollout data are collected using the current policy and stored in a sequence-aware on-policy rollout buffer. The recurrent states are maintained independently for each parallel environment. They are reset when an episode terminates, thereby preserving temporal information within each trajectory while preventing information leakage across episode boundaries. After each rollout is collected, the trajectories are split into contiguous subsequences of length L_{seq}, and each subsequence is paired with its initial recurrent state for truncated backpropagation through time.

Advantages are computed using GAE with discount factor γ and trace-decay factor λ. The value target for critic training is computed as

$$\hat{R}_t = \hat{A}_t + V_t,$$

where $V_t = V_{\phi_{\mathrm{old}}}(o_t, h_t)$ is the value estimate stored during rollout collection. During PPO optimization, mini-batches are formed at the sequence level rather than the individual-transition level to preserve recurrent temporal continuity. The policy and value networks are then updated using the PPO clipped objective and the value loss. Algorithm A.1 summarizes the rollout collection and recurrent PPO training procedure used in this work.

A.3. Full hyperparameter settings of PPO–VNS, DQN–VNS, and simulated annealing

For completeness, the full hyperparameter settings of simulated annealing, PPO–VNS, and DQN–VNS are summarized in Tables A.2–A.4.

Table A.2
Simulated annealing parameter settings.

Parameter	Value
Maximum iterations	1000
Initial temperature $\mathcal{T}_0$	100
Cooling coefficient α	0.99
Minimum temperature $\mathcal{T}_{\min}$	0.01

A.4. Marginal-gain analysis for sensitivity experiments

To provide quantitative support for the sensitivity analysis, we report the marginal-gain values between adjacent parameter levels. The marginal gain is computed based on the mean objective value across

Algorithm A.1 Rollout collection and recurrent PPO training.

TRAIN-RPPO($S, \Omega, \gamma, \lambda, \varepsilon, n_{\text{steps}}, K, L_{\text{seq}}, T, N_{\text{env}}$)
initialize parallel training environments $\{Env_i\}_{i=1}^{N_{\text{env}}}$
initialize recurrent actor–critic network with policy π_θ and value function V_ϕ
initialize recurrent states $(h_0^{(i)}, c_0^{(i)})$ for all environments
while total collected steps $< T$ **do**
  clear the on-policy rollout buffer
  for $t = 1, \dots, n_{\text{steps}}$ **do**
    for $i = 1, \dots, N_{\text{env}}$ **do**
      extract state features $s_t^{(i)}$ from the current solution
      form observation $o_t^{(i)} = [s_t^{(i)}, \text{onehot}(\omega_{t-1}^{(i)})]$
      compute policy distribution and value estimate using $(o_t^{(i)}, h_{t-1}^{(i)}, c_{t-1}^{(i)})$
      sample operator $\omega_t^{(i)} \sim \pi_\theta(\cdot \mid o_t^{(i)}, h_t^{(i)})$
      execute $\omega_t^{(i)}$ in Env_i and obtain $r_t^{(i)}$, $o_{t+1}^{(i)}$, and terminal flag $d_t^{(i)}$
      store rollout data:
      $\big(o_t^{(i)}, h_{t-1}^{(i)}, c_{t-1}^{(i)}, \omega_t^{(i)}, r_t^{(i)}, o_{t+1}^{(i)}, d_t^{(i)}, \log \pi_{\text{old}}(\omega_t^{(i)} \mid o_t^{(i)}, h_t^{(i)}), V_\phi(o_t^{(i)}, h_t^{(i)})\big)$
      update recurrent state to $(h_t^{(i)}, c_t^{(i)})$
      if $d_t^{(i)} = 1$ **then**
        reset recurrent state $(h_t^{(i)}, c_t^{(i)})$
      end if
    end for
  end for
  compute advantages $\hat{A}_t$ using GAE(γ, λ)
  compute value targets $\hat{R}_t = \hat{A}_t + V_t$, where $V_t = V_{\phi_{\text{old}}}(o_t, h_t)$
  normalize advantages within the rollout
  split each trajectory into contiguous subsequences of length L_{seq}
  attach the corresponding initial recurrent state to each subsequence
  for epoch $= 1, \dots, K$ **do**
    shuffle subsequences at the sequence level
    for all recurrent mini-batches **do**
      recompute action log-probabilities and value estimates on the subsequences
      compute the probability ratio
      $$\rho_t(\theta) = \frac{\pi_\theta(\omega_t \mid o_t, h_t)}{\pi_{\theta_{\text{old}}}(\omega_t \mid o_t, h_t)}$$
      maximize the PPO clipped objective
      $$L^{\text{CLIP}}(\theta) = \min\big(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\big)$$
      minimize the value loss
      $$\big(V_\phi(o_t, h_t) - \hat{R}_t\big)^2$$
      update θ and ϕ through backpropagation over the unrolled LSTM
    end for
  end for
end while
return π_θ, V_ϕ

Table A.3
PPO–VNS hyperparameter settings.

Hyperparameter	Value
Learning rate η	3×10^{-4}
PPO clipping ratio ε	0.2
Batch size	64
Number of rollout steps n_{steps}	128
Number of training epochs K	10
LSTM hidden size	256
Shared MLP layer widths	[128, 128]
Discount factor γ	0.99
GAE parameter λ	0.95
Total training steps T_{total}	1,000,000
Checkpoint interval	100,000
Reward weight for time term β_1	0.1
Reward weight for energy term β_2	0.05

Table A.4
DQN–VNS hyperparameter settings.

Hyperparameter	Value
Learning rate η	3×10^{-4}
Batch size	256
First hidden layer size	256
Second hidden layer size	256
Discount factor γ	0.99
Replay buffer size N	200,000
Learning starts	5,000
Target network update frequency C	1,000
Training frequency	4
Initial exploration rate ϵ_{init}	1.0
Final exploration rate ϵ_{final}	0.05
Exploration fraction	0.30
Total training steps	800,000
Checkpoint interval	100,000
Reward weight for time term β_1	0.1
Reward weight for energy term β_2	0.05

Table A.5
Marginal gains under different UAV fleet sizes.

Number of UAVs	Mean Value	95% CI	Marginal gain (%)
2	16458.96	953.80	–
3	20643.73	1662.06	25.43
4	22920.00	2303.05	11.03
5	25001.45	2939.49	9.08
6	26166.66	3518.96	4.66

Table A.6
Marginal gains under different UAV battery capacities.

Battery capacity	Mean Value	95% CI	Marginal gain (%)
1900	18047.69	722.74	–
2850	20129.80	779.17	11.54
4276	21456.00	863.16	6.59
5100	21582.52	860.23	0.59

the tested instances. The results for different UAV fleet sizes and battery capacities are presented in Tables A.5 and A.6, respectively. The 95% confidence intervals are also reported to indicate the variability of the mean objective value across instances.

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